MathScy: Domain-Specialized Mixture-of-Experts for AI-Assisted Mathematical Reasoning

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Abstract

We present MathScy, a modular system for AI-assisted mathematical reasoning that integrates domain-specialized Mixture-of-Experts (MoE) large language models with formal verification in the Lean 4 proof assistant. MathScy addresses fundamental limitations of current AI systems for mathematics—including scalability across mathematical domains, depth of reasoning, and verifiability of outputs—through three key contributions. First, we construct a domain-specialized MoE architecture via Branch-Train-Mix, training seven QLoRA expert modules on distinct mathematical subdomains (algebraic geometry, discrete mathematics, number theory, analysis, algebra, geometry & topology, and probability & statistics) plus a shared cross-domain expert from a DeepSeek-Math-7B base model, with sigmoid gating and auxiliary-loss-free load balancing. Second, we design a Self-Play Theorem Prover (STP) loop in which a conjecturer and a prover iteratively improve through mutual training, with conjectures filtered through a four-tier Lean 4 verification pipeline encompassing type-checking, triviality detection, counterexample search, and automated proof search. Third, we curate a large-scale dataset of 6.47 million mathematical entries across 154 arXiv categories and 15 domain groups, from which we extract 241,586 structured training examples via a combination of LLM-powered knowledge extraction and LaTeX theorem mining. We report results from all three pipeline phases: stable expert convergence across all domains on a single A100 GPU, a sigmoid router achieving 88.9% top-2 routing accuracy with near-uniform load balancing, and a conjecture generation pipeline producing 210 conjectures—of which 10 are judged as proved by LLM evaluators and 6 are flagged as publishable—with the STP loop achieving the highest quality scores (avg 0.735, max 0.980). A 12-task evaluation suite reveals that 54% of conjectures are genuinely novel (low ArXiv similarity), the optimal generation configuration uses temperature T=0.7 with 3 context entries, and STP quality in early rounds shows proved conjectures doubling each iteration, though extended analysis across 4 rounds reveals quality degradation complicated by API availability. A router gating ablation validates sigmoid over softmax gating with large effect size (Cohen's $d = 0.845$), and a multi-judge consensus study reveals weak positive inter-judge correlation ($r = 0.338$) with systematic scoring differences, underscoring the need for human evaluation. Under uniform LLM-judge evaluation, the MoE–API quality gap is 0.210, attributable primarily to instruction tuning rather than mathematical content quality. The current evaluation focuses on conjecture generation quality; problem-solving benchmarks (MATH, GSM8K) are planned as future work requiring GPU inference. The trained models, code, and LLM-verified conjectures are publicly available.[1]

Keywords: Mixture-of-Experts, mathematical reasoning, theorem proving, conjecture generation, Lean 4, autoformalization, self-play, domain specialization

1 Introduction

Mathematical reasoning—encompassing conjecture formulation, proof construction, and counterexample identification—lies at the heart of scientific and technological progress. From the

[1]Code: `https://github.com/ai4he/moexp` Models: `https://huggingface.co/ai4he/mathscy-moe-7b`

resolution of the Poincaré conjecture to the development of cryptographic protocols grounded in algebraic geometry, advances in mathematics have far-reaching consequences across the sciences [Hartshorne, 1977, Sturmfels, 2002]. Traditionally, the mathematical discovery process has been exclusively manual, requiring deep domain expertise and creative insight that is difficult to formalize or automate. Recent advances in artificial intelligence, particularly the Transformer architecture [Vaswani et al., 2017] and large language models (LLMs) built upon it—from GPT-2 [Radford et al., 2019] and BERT [Devlin et al., 2019] to GPT-3 [Brown et al., 2020] and instruction-tuned variants trained via reinforcement learning from human feedback [Ouyang et al., 2022]—have opened new possibilities for augmenting this process [Davies et al., 2021, Romera-Paredes et al., 2024, He, 2023], yet fundamental challenges remain.

Current AI systems for mathematical reasoning face four critical limitations. **Scalability**: mathematics encompasses a vast landscape of interconnected subdomains, each with its own vocabulary, proof techniques, and structural patterns; a model trained primarily on one area (e.g., combinatorics) may fail to transfer effectively to another (e.g., algebraic geometry) [Gururangan et al., 2020]. **Depth of reasoning**: while LLMs excel at pattern recognition and can generate plausible-looking mathematical text, they often struggle with the sustained logical reasoning required for complex proofs, particularly those requiring multiple intermediate lemmas or non-obvious proof strategies [OpenAI et al., 2024, Lewkowycz et al., 2022a]. **Verifiability**: LLM outputs are inherently probabilistic and prone to hallucination; without formal verification, there is no guarantee that a generated proof is correct or that a proposed conjecture is well-formed [Polu and Sutskever, 2020]. **Adaptability**: monolithic models require substantial retraining to incorporate new mathematical knowledge or adapt to emerging research areas [Ke et al., 2023].

We propose MathScy, a compositional and modular system designed to address these limitations through the integration of three key innovations:

1. **Domain-specialized Mixture-of-Experts architecture.** Rather than relying on a single general-purpose mathematical reasoner, MathScy employs a sparse MoE model with eight domain-specialized expert modules, each trained via Low-Rank Adaptation (LoRA) [Hu et al., 2022] on a distinct mathematical subdomain. The experts are constructed through the Branch-Train-Mix (BTM) paradigm [Sukhbaatar et al., 2024], which branches from a shared mathematical foundation (DeepSeek-Math-7B [Shao et al., 2024]), trains each branch on domain-specific data in embarrassingly parallel fashion, and mixes the specialized modules into a unified MoE architecture with sigmoid gating and auxiliary-loss-free load balancing [DeepSeek-AI et al., 2024b]. This design enables efficient scaling across mathematical domains while preserving within-domain depth.

2. **Self-Play Theorem Prover with formal verification.** Inspired by the expert iteration paradigm [Anthony et al., 2017, Silver et al., 2017] and recent advances in neural theorem proving [Polu et al., 2023, Ren et al., 2025], MathScy implements a self-play loop in which a conjecturer model and a prover model iteratively improve through mutual training. Generated conjectures are autoformalized into Lean 4 [de Moura and Ullrich, 2021] and subjected to a four-tier verification pipeline: type-checking, triviality detection via `aesop`, counterexample search, and automated proof attempts using Mathlib tactics [The mathlib Community, 2020]. Verification outcomes provide a noise-free training signal that grounds the self-play loop in formal correctness.

3. **Large-scale structured mathematical dataset.** We curate a corpus of 6.47 million mathematical entries spanning 154 arXiv categories and 15 domain groups, from which we extract 241,586 structured training examples through a combination of Gemini-powered knowledge extraction and LaTeX theorem mining from arXiv source files. The dataset is explicitly organized by mathematical type and domain, enabling stratified training of domain-specialized experts.

The system focuses on current research in applied, combinatorial, and computational algebraic geometry—areas where computational experimentation and AI-assisted discovery have particular potential to advance mathematical knowledge [Cox et al., 2015, Sommese and Wampler, 2005]. Algebraic geometry's rich interplay between algebraic structures and geometric intuition, combined with the computability of many fundamental invariants (Gröbner bases, Hilbert functions, intersection multiplicities), makes it an ideal testbed for AI-assisted mathematical reasoning.

This paper makes the following contributions:

- We present the first domain-specialized MoE architecture for mathematical reasoning, demonstrating that parameter-efficient expert modules trained on distinct mathematical subdomains can be effectively combined through learned routing.
- We design a complete self-play theorem proving pipeline that integrates conjecture generation, autoformalization, and formal verification in Lean 4, creating a closed loop between informal mathematical intuition and machine-checkable proofs.
- We construct one of the largest structured mathematical knowledge bases assembled for training domain-specialized models, with explicit type and domain annotations enabling stratified expert training.
- We provide a comprehensive experimental framework for evaluating AI-assisted mathematical reasoning systems, including a 12-task conjecture quality evaluation suite and six ablation studies covering generation strategy, temperature, context, routing, STP iteration, and cross-domain transfer.
- We release all code, trained model weights, and proved conjectures as open-source resources to facilitate reproducibility and future research.[2]

The remainder of this paper is organized as follows. Section 2 surveys related work across the relevant research areas. Section 3 establishes the mathematical foundations underlying our approach. Section 4 presents the MathScy system architecture and training methodology in detail. Section 5 describes the dataset construction pipeline. Section 6 outlines the experimental protocol. Section 7 presents results from all three pipeline phases, including comprehensive conjecture evaluation (Section 7.8) and ablation studies (Section 7.9). Section 8 discusses implications and limitations. Section 9 concludes.

2 Related Work

Our work lies at the intersection of several active research areas: mixture-of-experts architectures, large language models for mathematical reasoning, formal theorem proving, conjecture generation, mathematical dataset construction, domain specialization, and self-play for reasoning. We survey each area in turn, situating MathScy within the broader landscape.

2.1 Mixture-of-Experts Models

The Mixture-of-Experts (MoE) paradigm has its roots in the seminal work of Jacobs et al. [1991], who proposed a system of expert networks governed by a gating mechanism that partitions the input space, enabling different experts to specialize on different subsets of the data. This foundational idea—that a collection of specialized sub-models can outperform a single monolithic model—has undergone a significant resurgence in the era of large-scale Transformers. The key

[2]Source code and pipeline: `https://github.com/ai4he/moexp`. Pre-trained MoE adapters and router: `https://huggingface.co/ai4he/mathscy-moe-7b`.

insight driving modern MoE research is that sparse conditional computation allows models to scale parameter counts dramatically while keeping per-token computational cost tractable, a property first demonstrated at scale by the Sparsely-Gated Mixture-of-Experts layer of Shazeer et al. [2017], which interposed a trainable gating network between Transformer layers and a large pool of feed-forward experts, activating only the top-k experts per token.

The subsequent evolution of MoE architectures has been defined by innovations in routing, load balancing, and scaling efficiency. Lepikhin et al. [2021] introduced GShard, which scaled MoE to 600 billion parameters for multilingual machine translation with a top-2 gating strategy and random auxiliary routing to a second expert, demonstrating the viability of MoE for cross-lingual generalization. The Switch Transformer [Fedus et al., 2022] simplified this design by employing top-1 routing—dispatching each token to a single expert—and introducing differentiable load-balancing auxiliary losses alongside capacity factors to prevent expert collapse, enabling models to scale to over one trillion parameters with reduced communication overhead. Concurrently, GLaM [Du et al., 2022] demonstrated that a 1.2-trillion-parameter MoE model with 64 experts per layer could match GPT-3 quality while using only one-third of the energy for training, and Riquelme et al. [2021] showed that MoE principles extend effectively to vision transformers. Zhou et al. [2022] provided a comprehensive survey of MoE methods, systematically categorizing advances in gating mechanisms, training strategies, and applications. ST-MoE [Zoph et al., 2022] further refined the design space by introducing stable training techniques and encoder-decoder MoE models that achieved state-of-the-art performance on reasoning benchmarks. These works established auxiliary losses as the standard mechanism for encouraging balanced expert utilization, though at the cost of introducing a hyperparameter whose magnitude must be carefully tuned: too large an auxiliary loss degrades task performance, while too small a loss permits routing degeneration.

More recently, a new generation of MoE models has pushed beyond auxiliary-loss-based load balancing. The DeepSeekMoE architecture [Dai et al., 2024] proposed fine-grained expert segmentation combined with shared experts, isolating common knowledge in a subset of always-active experts while routing specialized tokens to domain-specific ones. DeepSeek-V2 [DeepSeek-AI et al., 2024a] extended this with Multi-head Latent Attention and further refined the expert-routing paradigm. Most relevant to our work, DeepSeek-V3 [DeepSeek-AI et al., 2024b] introduced an auxiliary-loss-free load balancing strategy that replaces explicit auxiliary losses with a dynamic bias term added to affinity scores during top-K routing: expert load is monitored at each training step, and the bias is adjusted to penalize overloaded experts and promote underloaded ones. Critically, DeepSeek-V3 also adopted sigmoid gating—computing affinity scores via the sigmoid function and normalizing among selected experts—rather than the conventional softmax over all experts, yielding more flexible multi-expert activation patterns. This combination of sigmoid gating with loss-free balancing achieves superior performance to auxiliary-loss methods, and directly informs the gating design in MathScy.

The open-source MoE ecosystem has also matured substantially. Mixtral 8×7B [Jiang et al., 2024] demonstrated that a sparse mixture of eight experts per layer, with top-2 routing and 12.9B active parameters out of 46.7B total, can match or exceed models several times its effective size, including Llama 2 70B. OLMoE [Muennighoff et al., 2024] provided a fully open, reproducible MoE baseline with 7B total parameters and 1B active, trained on 5 trillion tokens, accompanied by open-source weights, training data, and code—establishing important baselines for MoE research and demonstrating that even modestly-sized MoE models can achieve competitive performance through careful data curation and training.

Of particular importance to our architectural design is the Branch-Train-Mix (BTX) framework of Sukhbaatar et al. [2024], which proposes a three-stage recipe for constructing domain-specialized MoE models: (i) branching from a seed pretrained model by copying it, (ii) training each copy independently on a domain-specific data subset in embarrassingly parallel fashion, and (iii) mixing the specialized experts into MoE layers with averaged non-expert parameters,

followed by MoE fine-tuning to learn token-level routing. BTX generalizes both Branch-Train-Merge [Li et al., 2022] (which lacks learned routing) and sparse upcycling (which omits independent expert training), demonstrating superior accuracy and efficiency on mathematics and code domains without catastrophic forgetting. MathScy adopts the BTX paradigm but replaces full-model branching with parameter-efficient LoRA expert modules, enabling us to train eight domain-specialized mathematical experts from a shared DeepSeek-Math base with substantially reduced memory and compute requirements.

2.2 Large Language Models for Mathematical Reasoning

The application of large language models to mathematical reasoning has progressed rapidly from early demonstrations of arithmetic and word-problem solving to systems approaching competition-level mathematics. Early benchmarks such as the DeepMind Mathematics Dataset [Saxton et al., 2019] and the ARC challenge [Clark et al., 2018] revealed both the promise and limitations of neural approaches to mathematical and abstract reasoning tasks. The scaling paradigm established by T5 [Raffel et al., 2020], PaLM [Chowdhery et al., 2023], Gopher [Rae et al., 2022], and the Chinchilla scaling laws [Hoffmann et al., 2022] demonstrated that model and data scale are critical determinants of mathematical capability. The LLaMA family [Touvron et al., 2023a,b] democratized access to capable base models, while Galactica [Taylor et al., 2022] showed that pretraining on scientific corpora yields strong performance on mathematical tasks. A pivotal advance was the discovery by Wei et al. [2022] that chain-of-thought (CoT) prompting—providing a few exemplar reasoning chains as demonstrations—elicits multi-step reasoning capabilities that emerge with sufficient model scale (approximately 100B parameters), dramatically improving performance on arithmetic, commonsense, and symbolic reasoning benchmarks. This emergent property of large models underscored the importance of scale and prompted the development of mathematics-specialized architectures.

Minerva [Lewkowycz et al., 2022a], based on PaLM, demonstrated that continued pretraining on 118GB of scientific papers from the arXiv, combined with few-shot CoT prompting and majority voting, could achieve state-of-the-art performance on quantitative reasoning tasks spanning mathematics, physics, and other STEM domains. Follow-up work by Lewkowycz et al. [2022b] extended Minerva's approach to broader scientific reasoning tasks beyond mathematics. However, Minerva's answers could not be automatically verified, as the model operates entirely in natural language without recourse to formal grounding. GPT-4 [OpenAI et al., 2024] further raised the ceiling on mathematical reasoning, achieving strong performance on competition-level benchmarks including the AMC and AIME, though its closed-source nature limits its utility as a research platform.

The open-source mathematical reasoning community has produced a series of increasingly capable models. Llemma [Azerbayev et al., 2024], continuing pretraining Code Llama on the Proof-Pile-2 mixture of scientific papers, mathematical web text, and formal proof data, outperformed all open base models on the MATH benchmark and demonstrated capability in both tool use and formal theorem proving without task-specific fine-tuning. DeepSeekMath [Shao et al., 2024] pushed the boundaries further by leveraging a meticulously engineered web data selection pipeline to harvest high-quality mathematical content from Common Crawl, and introduced Group Relative Policy Optimization (GRPO)—a memory-efficient variant of PPO—to enhance mathematical reasoning through reinforcement learning, achieving 51.7% on the MATH benchmark at the 7B scale. InternLM-Math [Ying et al., 2024b] contributed a chain-of-thought reasoning framework integrated with tool use and formal verification capabilities, while Qwen2.5-Math [Yang et al., 2024] established the current state of the art among open models by supporting both CoT and Tool-Integrated Reasoning (TIR) across multiple languages.

The integration of external computational tools with natural language reasoning has emerged as a powerful paradigm for mathematical problem solving. ToRA [Gou et al., 2024] introduced Tool-integrated Reasoning Agents that interleave natural language reasoning steps with calls to

computation libraries and symbolic solvers, trained via imitation learning on GPT-4-generated interactive trajectories. ToRA demonstrated that even modest-scale models (34B) can outperform GPT-4's CoT performance on competition-level benchmarks when equipped with appropriate tool interfaces. Process-level verification [Lightman et al., 2024] further enhanced reasoning reliability by training reward models to verify individual reasoning steps rather than only final answers, enabling more fine-grained feedback during both training and inference. Math-Shepherd [Wang et al., 2024b] extended process supervision with automatic step-level reward labeling, enabling scalable reward model training. More recently, DeepSeek-R1 [DeepSeek-AI et al., 2025] demonstrated that reinforcement learning alone can induce sophisticated mathematical reasoning behaviors, including self-verification and chain-of-thought, without supervised fine-tuning. AlphaGeometry [Trinh et al., 2024] combined a neural language model with a symbolic deduction engine to solve Olympiad-level geometry problems, approaching human gold-medalist performance. The MUSTARD benchmark [Huang et al., 2024] was introduced to evaluate multi-step textual reasoning with distractors, providing a more rigorous test of mathematical reasoning robustness.

MathScy builds upon these advances by combining domain-specialized MoE architecture with mathematical reasoning, hypothesizing that different areas of mathematics—from algebraic geometry to combinatorics to probability theory—benefit from dedicated expert modules rather than a single general-purpose mathematical reasoner. Our approach complements the reasoning enhancements of CoT, GRPO, and tool integration by introducing an orthogonal axis of improvement: structural specialization at the architecture level.

2.3 Formal Theorem Proving with Large Language Models

The intersection of large language models and formal theorem proving represents one of the most promising avenues for achieving verifiable mathematical reasoning. Earlier work on premise selection for theorem proving by Ferreira and Freitas [2020] demonstrated that natural language processing techniques could identify relevant premises from large libraries, and TacticToe [Gauthier et al., 2021] applied Monte Carlo tree search to tactic-based proof search in HOL4. The pioneering GPT-f system [Polu and Sutskever, 2020] demonstrated that Transformer-based language models could generate novel proofs in the Metamath formal system, contributing 23 shortened proofs that were accepted into the main Metamath library—the first instance of a deep learning system contributing proofs adopted by a formal mathematics community. Subsequent work by Polu et al. [2023] on formal mathematics statement curriculum learning showed that expert iteration with a curriculum of progressively harder statements could substantially improve proof search, establishing a paradigm in which language models are iteratively trained on their own successful proofs.

The development of interactive proof assistants, particularly Lean 4 and its mathematical library Mathlib, has created a rich ecosystem for neural theorem proving. Early efforts in Lean-based autoformalization by Jiang et al. [2022] demonstrated the feasibility of translating informal mathematical statements into Lean, while Wu et al. [2022] showed that large language models can autoformalize mathematical statements at scale with surprisingly high accuracy. Zheng et al. [2024] explored verification techniques within the Lean ecosystem, and recent progress surveyed by Ying et al. [2024a] highlights the rapid maturation of Lean-based proving tools. LeanDojo [Yang et al., 2023b] provided a comprehensive toolkit for extracting training data from Lean repositories and interacting with the Lean prover programmatically, enabling the training of ReProver—a retrieval-augmented theorem prover that selects relevant premises from the library to guide proof search. LeanCopilot [Song et al., 2024] brought large language models directly into the Lean development environment as a copilot, suggesting tactics and premises in real time. LeanAnime [Yang et al., 2023a] introduced animated visualization techniques to make formal proofs more accessible. The LEGO-Prover [Wang et al., 2024a] introduced a modular approach to proof construction, maintaining a growing library of verified lemmas that can be

retrieved and composed to prove new theorems, advancing the state of the art on the miniF2F benchmark [Zheng et al., 2022].

The most dramatic recent advance in formal theorem proving is AlphaProof [Hubert et al., 2025], which coupled a pretrained language model with the AlphaZero reinforcement learning algorithm to achieve silver-medal performance at the 2024 International Mathematical Olympiad, solving four out of six problems. AlphaProof guarantees correctness by verifying every logical step using the Lean proof assistant, and employs test-time reinforcement learning to enable deep, problem-specific adaptation. The DeepSeek-Prover series has also made significant contributions: DeepSeek-Prover [Xin et al., 2024a] introduced Monte Carlo tree search for proof generation in Lean 4, DeepSeek-Prover-V1.5 [Xin et al., 2024b] refined this approach with improved training data curation and proof search strategies, and DeepSeek-Prover-V2 [Ren et al., 2025] advanced formal reasoning through subgoal decomposition, achieving 88.9% on miniF2F-test with the 671B variant. Draft, Sketch, and Prove [Jiang et al., 2023] and Baldur [First et al., 2023] explored complementary strategies of proof sketching and whole-proof generation, respectively. NaturalProver [Welleck et al., 2022] generated proofs by conditioning on natural language references and definitions, bridging the gap between informal and formal reasoning. Theorem-Llama [Chen et al., 2024] fine-tuned the LLaMA architecture specifically for theorem proving, demonstrating that open-weight models can be adapted into capable provers. MagnusHammer [Mikuła et al., 2024] advanced premise selection through transformer-based retrieval, significantly improving tactic-based proof search by identifying the most relevant library lemmas.

MathScy integrates formal verification as a core component of its Self-Play Theorem Prover loop, using Lean 4 autoformalization to translate naturally-expressed conjectures into formally verifiable statements. Unlike systems that focus exclusively on formal proof search, our approach treats autoformalization as a bridge between the conjecture generation capabilities of domain-specialized experts and the verification guarantees of formal proof assistants.

2.4 Conjecture Generation and Mathematical Discovery

The automation of conjecture generation—producing novel, non-trivial mathematical hypotheses—represents a fundamentally different challenge from theorem proving, requiring creative exploration of mathematical structure rather than verification of known claims. A watershed moment for AI-driven mathematical discovery came with Davies et al. [2021], who demonstrated that machine learning could guide human mathematical intuition by identifying previously unknown patterns in knot theory and representation theory. By training neural networks to predict relationships between mathematical invariants and then interpreting the learned representations, the collaboration led to the discovery of a new geometric quantity and new connections between Bruhat intervals and Kazhdan–Lusztig polynomials.

Fawzi et al. [2022] introduced AlphaTensor, which cast the discovery of efficient matrix multiplication algorithms as a reinforcement learning problem, discovering algorithms that surpassed Strassen's 50-year-old method for certain matrix sizes. FunSearch [Romera-Paredes et al., 2024], published in Nature, represented the first instance of an LLM-based system making genuine new mathematical discoveries, discovering new constructions of large cap sets exceeding previously known bounds by searching the space of programs rather than solutions. Poesia et al. [2024] proposed learning formal mathematics from intrinsic motivation, jointly training a conjecturer and prover in a self-play framework that bootstraps from axioms alone. Wagner [Wagner, 2021] approached conjecture generation through reinforcement learning for counterexample search, and He [2023] surveyed the emerging applications of machine learning to pure mathematics.

MathScy contributes to this landscape by coupling domain-specialized expert models with a self-play loop that jointly trains a conjecturer and a prover, with domain specialization enabling conjectures grounded in specific mathematical vocabularies and structural patterns.

2.5 Dataset Construction for Mathematical AI

The quality and coverage of training data have emerged as decisive factors in the performance of mathematical reasoning systems. The MATH dataset [Hendrycks et al., 2021] established a foundational benchmark of 12,500 competition-level problems with detailed solutions. OpenWebMath [Paster et al., 2024] extracted 14.7 billion tokens of mathematical web content from Common Crawl, while MathPile [Wang et al., 2023b] assembled a diverse 9.5-billion-token corpus from textbooks, arXiv papers, and mathematical websites. For formal mathematics, ProofNet [Azerbayev et al., 2023] provided undergraduate-level formal proving benchmarks in Lean 4, NaturalProofs [Welleck et al., 2021] assembled 30,000 theorem-proof pairs from mathematical repositories, and the Goedel dataset [Lin et al., 2025] produced 1.64 million formalized statements through synthetic data generation.

Beyond mathematics-specific datasets, broader evaluation frameworks such as HumanEval [Chen et al., 2021] for code generation, BigCodeBench [Zhuo et al., 2024] for compositional coding tasks, and HELM [Liang et al., 2023] for holistic language model evaluation have established methodological standards for rigorous model assessment that inform our evaluation design.

Our dataset of 6.47 million mathematical entries across 15 domains, explicitly structured around mathematical type and domain, represents one of the largest structured collections assembled for training domain-specialized models, complementing the pretraining-focused corpora (OpenWebMath, MathPile) and formal verification datasets (ProofNet, Goedel).

2.6 Domain Specialization and Parameter-Efficient Fine-Tuning

The effectiveness of domain-adaptive pretraining was established by Gururangan et al. [2020], who showed that a second phase of pretraining on in-domain text yields consistent performance gains across diverse domains. Parameter-efficient fine-tuning (PEFT) methods make multi-domain specialization tractable: prompt tuning [Lester et al., 2021] demonstrated that learning soft prompts can adapt frozen models to downstream tasks with minimal parameters, LoRA [Hu et al., 2022] represents weight updates as low-rank decompositions, QLoRA [Dettmers et al., 2023] enables fine-tuning through 4-bit quantized base models, ReFT [Wu et al., 2024] introduced representation fine-tuning that intervenes on hidden representations rather than weights, and DoRA [Liu et al., 2024] refines this by decomposing weight matrices into magnitude and direction components. The adapter framework of Houlsby et al. [2019] pioneered task-specific bottleneck modules with minimal parameter overhead. Scialom et al. [2022] provided a comprehensive survey of fine-tuning strategies for language models, while Gupta et al. [2023] addressed the challenge of continual learning in the context of domain adaptation, proposing methods to prevent catastrophic forgetting when models are sequentially adapted to new domains.

In MathScy, each expert consists of LoRA adapters specialized to a mathematical domain, while the shared base parameters of DeepSeek-Math-7B encode general mathematical knowledge—enabling eight domain experts to be trained with a fraction of the memory required by full-model branching.

2.7 Self-Play and Reinforcement Learning for Reasoning

Self-play reinforcement learning, as demonstrated by AlphaZero [Silver et al., 2017], achieves superhuman performance through agents learning by competing against themselves. Expert Iteration [Anthony et al., 2017] formalized this as a dual-process framework, and Proximal Policy Optimization [Schulman et al., 2017] provided a stable policy gradient algorithm that has become the standard for reinforcement learning from human feedback in language models. Self-Instruct [Wang et al., 2023a] demonstrated that language models can generate their own instruction-following training data, enabling self-improvement without human annotation. Kadavath et al. [2022] investigated language models' capacity for self-knowledge and calibrated self-evaluation,

showing that models can assess the correctness of their own outputs—a capability central to self-play frameworks. STaR [Zelikman et al., 2022] introduced self-taught reasoning for language models, and ReST [Gulcehre et al., 2023] generalized this through an expectation-maximization framework. BIG-bench [Srivastava et al., 2023] provided a comprehensive benchmark suite for evaluating diverse reasoning capabilities, including mathematical reasoning tasks that inform self-play evaluation. Singh et al. [2024] demonstrated that iterative self-training with correctness feedback surpasses human-data fine-tuning for mathematical reasoning. In formal mathematics, Polu et al. [2023] showed that expert iteration with curriculum learning dramatically improves proof search.

MathScy extends the self-play paradigm by introducing domain-specialized experts into the conjecturer-prover loop, ensuring that conjectures remain mathematically meaningful within their respective fields.

3 Background and Preliminaries

We establish the mathematical foundations underlying MathScy's architecture. Our system integrates four key technical pillars: sparse Mixture-of-Experts models for domain-specialized reasoning, parameter-efficient fine-tuning via low-rank adaptation, formal verification through dependent type theory, and self-play theorem proving for iterative improvement.

3.1 Mixture-of-Experts Models

The Mixture-of-Experts (MoE) paradigm [Jacobs et al., 1991] enables conditional computation by routing each input to a subset of specialized sub-networks, thereby decoupling model capacity from per-example computational cost.

3.1.1 Standard MoE Formulation

Consider an input representation $\mathbf{x} \in \mathbb{R}^d$ at a given layer of a transformer network. An MoE layer replaces the monolithic feed-forward network (FFN) with N expert networks $\{E_i\}_{i=1}^N$, each $E_i : \mathbb{R}^d \to \mathbb{R}^d$, and a gating function $g : \mathbb{R}^d \to \mathbb{R}^N$. The layer output is the gated combination:

$$\mathbf{y} = \sum_{i=1}^{N} g_i(\mathbf{x}) \cdot E_i(\mathbf{x}), \tag{1}$$

where $g_i(\mathbf{x})$ denotes the i-th component of the gating vector. Each expert E_i is typically a multi-layer perceptron with SwiGLU activation [Shazeer et al., 2017]:

$$E_i(\mathbf{x}) = \big(\sigma(W_1^{(i)}\mathbf{x}) \odot W_3^{(i)}\mathbf{x}\big)W_2^{(i)}, \tag{2}$$

where $W_1^{(i)}, W_3^{(i)} \in \mathbb{R}^{d_{\text{ff}} \times d}$, $W_2^{(i)} \in \mathbb{R}^{d \times d_{\text{ff}}}$, $\sigma(\cdot)$ is the SiLU nonlinearity, and $\odot$ denotes the Hadamard product.

3.1.2 Top-k Routing

The seminal sparse gating mechanism [Shazeer et al., 2017] computes gate values via a learned linear projection followed by a sparsified softmax. Let $W_g \in \mathbb{R}^{N \times d}$ be the gating weight matrix. The gating function is:

$$g(\mathbf{x}) = \mathrm{TopK}\big(\mathrm{softmax}(W_g\mathbf{x}),\ k\big), \tag{3}$$

where $\mathrm{TopK}(\cdot, k)$ retains the k largest entries and zeros out the rest. Letting $\mathbf{s} = \mathrm{softmax}(W_g \mathbf{x}) \in \Delta^{N-1}$ and $\mathcal{T}(\mathbf{x}) = \mathrm{argtopk}(\mathbf{s}, k)$:

$$g_i(\mathbf{x}) = \begin{cases} \frac{s_i}{\sum_{j \in \mathcal{T}(\mathbf{x})} s_j} & \text{if } i \in \mathcal{T}(\mathbf{x}), \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

The Switch Transformer [Fedus et al., 2022] demonstrated that $k = 1$ suffices for strong performance while maximizing computational efficiency.

3.1.3 Sigmoid Routing with Auxiliary-Loss-Free Balancing

DeepSeek-V3 [DeepSeek-AI et al., 2024b] introduced an alternative gating mechanism replacing the softmax with independent sigmoid activations:

$$g_i(\mathbf{x}) = \sigma\big(\mathbf{w}_i^\top \mathbf{x} + b_i\big), \tag{5}$$

where $\mathbf{w}_i \in \mathbb{R}^d$ is the gating weight vector, $b_i \in \mathbb{R}$ is a learned bias, and $\sigma(z) = (1+e^{-z})^{-1}$ is the logistic sigmoid. The bias terms serve as auxiliary-loss-free balancing knobs: during training, letting $c_i^{(t)}$ denote the cumulative token count for expert i at step t:

$$b_i^{(t+1)} = b_i^{(t)} - \gamma \cdot \mathrm{sign}\left(c_i^{(t)} - \frac{1}{N} \sum_{j=1}^{N} c_j^{(t)} \right), \tag{6}$$

where $\gamma > 0$ is a small step size, decoupling load balancing from the training loss landscape.

3.1.4 Load Balancing via Auxiliary Loss

When explicit balancing is employed [Shazeer et al., 2017, Fedus et al., 2022], the training loss is augmented with:

$$f_i = \frac{1}{T} \sum_{t=1}^{T} \mathbb{1}\big[i \in \mathcal{T}(\mathbf{x}_t)\big], \qquad P_i = \frac{1}{T} \sum_{t=1}^{T} s_i(\mathbf{x}_t), \tag{7}$$

$$\mathcal{L}_{\text{aux}} = \alpha \cdot N \cdot \sum_{i=1}^{N} f_i \cdot P_i, \tag{8}$$

with hyperparameter $\alpha > 0$, yielding total objective $\mathcal{L} = \mathcal{L}_{\text{task}} + \mathcal{L}_{\text{aux}}$.

3.1.5 Shared Expert and Branch-Train-Mix

We augment the MoE layer with a shared expert E_{shared} that processes every token unconditionally [DeepSeek-AI et al., 2024b]:

$$\mathbf{y} = E_{\text{shared}}(\mathbf{x}) + \sum_{i=1}^{N} g_i(\mathbf{x}) \cdot E_i(\mathbf{x}). \tag{9}$$

The Branch-Train-Mix paradigm [Sukhbaatar et al., 2024] constructs MoE models in three phases: **Branch**—create N copies of a pretrained model $\mathcal{M}_0$ with parameters $\theta_0 = (\theta_{\text{attn}}, \theta_{\text{ffn}})$; **Train**—continue pretraining each $\mathcal{M}_i$ on domain corpus $\mathcal{D}_i$, yielding θ_i; **Mix**—install each branch's FFN as a routed expert and average attention parameters:

$$\theta_{\text{attn}}^{(\text{MoE})} = \frac{1}{N} \sum_{i=1}^{N} \theta_{\text{attn}}^{(i)}. \tag{10}$$

3.2 Low-Rank Adaptation

3.2.1 LoRA Formulation

For a pretrained weight matrix $W_0 \in \mathbb{R}^{d\times k}$, LoRA [Hu et al., 2022] decomposes the additive update as:

$$W = W_0 + \Delta W = W_0 + \frac{\alpha}{r} BA, \tag{11}$$

where $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, and $r \ll \min(d,k)$. At initialization, $A \sim \mathcal{N}(0,\sigma^2)$ and $B = \mathbf{0}$, ensuring $\Delta W = \mathbf{0}$ initially. The number of trainable parameters per layer is $r(d+k)$, a reduction by factor $\min(d,k)/r$ from full fine-tuning.

3.2.2 QLoRA

QLoRA [Dettmers et al., 2023] quantizes frozen base weights to 4-bit NormalFloat (NF4) precision—choosing quantization levels as quantiles of $\mathcal{N}(0,1)$:

$$\Phi(q_j) = \frac{j}{2^b}, \quad j = 0, 1, \ldots, 2^b, \tag{12}$$

where Φ is the standard normal CDF. Double quantization further reduces metadata overhead, and paged optimizers manage CPU-GPU memory transfer.

3.2.3 LoRA-MoE Construction

In MathScy, each routed expert is a LoRA adapter applied to the base FFN:

$$\mathbf{y} = \mathrm{FFN}(W_0;\ \mathbf{x}) + \sum_{i=1}^{N} g_i(\mathbf{x}) \cdot \frac{\alpha}{r}\big(B_i A_i\big)\mathbf{x}, \tag{13}$$

reducing trainable parameters from $N \cdot dk$ (full MoE) to $N \cdot r(d+k)$.

3.3 Formal Verification in Lean 4

Lean 4 [de Moura and Ullrich, 2021] is based on the Calculus of Inductive Constructions (CIC), a dependent type theory where the Curry-Howard correspondence identifies propositions with types and proofs with terms:

$$\underbrace{P \text{ is true}}_{\text{logic}} \quad \longleftrightarrow \quad \underbrace{\exists\, t \text{ such that } t : P}_{\text{type theory}}. \tag{14}$$

Lean 4's kernel type-checker algorithmically verifies that a proof term t has the claimed type P, reducing proof verification to type checking—a decidable procedure in CIC. Mathlib [The mathlib Community, 2020] provides over 170,000 formalized definitions and theorems spanning algebra, analysis, topology, and combinatorics. The formalization of algebraic structures in Mathlib draws on foundational treatments of commutative algebra [Eisenbud, 1995], enabling formal reasoning about rings, ideals, and modules that underpin algebraic geometry.

Autoformalization—translating informal mathematical statements into formal Lean 4 terms—must satisfy: (1) type-correctness ($\hat{s}$ passes the kernel type-checker), (2) semantic fidelity ($\hat{s}$ faithfully represents the content of s), and (3) provability preservation (true theorems yield provable formalizations).

3.4 Self-Play Theorem Proving

Inspired by expert iteration [Anthony et al., 2017, Silver et al., 2017], we employ a self-play loop alternating between policy improvement via search and policy distillation:

$$\pi_n^* = \mathrm{Search}(\pi_{\theta_n}, \mathcal{V}), \tag{15}$$

$$\theta_{n+1} = \arg\min_{\theta} \ \mathbb{E}_{x\sim\mathcal{D}}\big[D_{\mathrm{KL}}(\pi_n^*(\cdot \mid x) \,\|\, \pi_\theta(\cdot \mid x))\big], \tag{16}$$

where $\mathcal{V}$ is instantiated as the Lean 4 type-checker, providing a sparse but noise-free training signal [Polu et al., 2023].

The conjecturer-prover loop generates candidates $s \sim \mathcal{C}_\phi(\cdot \mid c)$, autoformalizes them, filters through a quality criterion:

$$Q(\hat{s}) = \mathbb{1}[\hat{s} \text{ type-checks}] \cdot \mathbb{1}[\texttt{aesop}(\hat{s}) = \texttt{fail}] \cdot \mathbb{1}[\texttt{slim_check}(\hat{s}) = \texttt{pass}], \tag{17}$$

and trains both models on successful proof pairs, creating a virtuous cycle of improvement.

4 Methodology

We present MathScy, an integrated system for AI-assisted mathematical reasoning that combines domain-specialized Mixture-of-Experts (MoE) language models with a self-play theorem proving loop and formal verification in Lean 4. The system supports conjecture formulation, proof construction, and counterexample identification, with a focus on applied, combinatorial, and computational algebraic geometry. In this section, we describe each component in detail.

4.1 System Architecture Overview

MathScy comprises four tightly integrated subsystems: (1) a data curation pipeline that harvests and structures mathematical knowledge from the arXiv corpus; (2) a domain-specialized MoE language model built via the Branch-Train-Mix paradigm [Sukhbaatar et al., 2024]; (3) a Self-Play Theorem Prover (STP) loop that iteratively improves both conjecture generation and proof search; and (4) a Lean 4 autoformalization and verification pipeline [de Moura and Ullrich, 2021] that provides rigorous correctness guarantees.

The base language model is `deepseek-math-7b-base` [Shao et al., 2024], a 7-billion parameter transformer pretrained on 120 billion math-related tokens. We augment this model with N=8 domain-specialized LoRA experts [Hu et al., 2022] plus one shared expert that is always active, following the DeepSeekMoE design philosophy [DeepSeek-AI et al., 2024b]. A learned sigmoid router with auxiliary-loss-free load balancing dispatches each input token to the top-k (k=2) most relevant domain experts (see Figure 1). The prover component is built on DeepSeek-Prover-V2-7B [Ren et al., 2025], fine-tuned via LoRA on proof data generated during the STP loop.

All training is conducted using QLoRA [Dettmers et al., 2023] with 4-bit NF4 quantization and double quantization, enabling the full pipeline to run within a 72-hour allocation on a DGX node equipped with 8× NVIDIA A100 or H100 GPUs (80 GB each). Distributed training employs the DeepSpeed library [Rasley et al., 2020] with ZeRO Stage 2 [Rajbhandari et al., 2020] for optimizer state and gradient partitioning, drawing on parallelism techniques pioneered by Megatron-LM [Shoeybi et al., 2020] for efficient multi-GPU training of large transformer models.

4.2 Data Collection and Curation

Our training corpus is derived from a deduplicated collection of 6,472,665 mathematical entries extracted from arXiv papers, spanning 154 arXiv categories. The dataset encompasses diverse mathematical statement types, including theorems (∼1.24M), lemmas (∼1.39M), propositions (∼910K), definitions (∼723K), remarks (∼769K), and conjectures (∼43.5K).

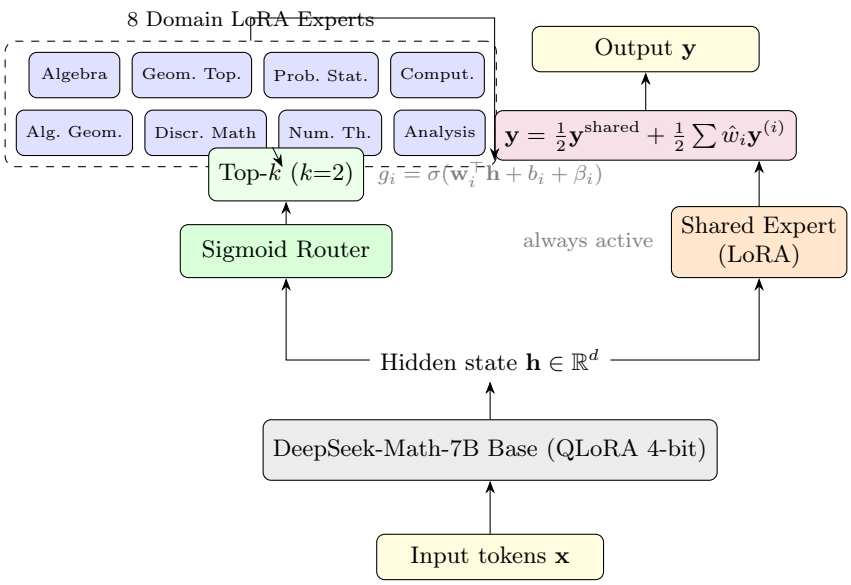

Figure 1: Architecture of the MathScy MoE model. Input tokens are processed by the QLoRA-quantized DeepSeek-Math-7B base model, producing hidden states $\mathbf{h}$. A sigmoid router computes per-expert gating scores and selects the top-k (k=2) domain experts. The final output combines contributions from the selected domain experts and the always-active shared expert in equal proportion (Eq. 20).

Stratified sampling. To ensure balanced domain representation, we construct a stratified sample of 3,400 entries via reservoir sampling, drawing 100 entries from each of 34 primary mathematics categories. This sample serves as the seed for both structured knowledge extraction and ArXiv LaTeX theorem mining.

Structured knowledge extraction. We employ Gemini-2.0-Flash in JSON mode with temperature τ=0.3 to extract structured representations. Each entry is processed into a JSON schema containing: formal statement, informal description, key concepts, prerequisites, mathematical domain, related theorems, and potential generalizations. A multi-strategy JSON parser with LaTeX-aware backslash fixing handles Gemini's occasional escape issues, achieving a 72.4% successful parse rate across 3,319 processed entries.

ArXiv LaTeX theorem extraction. In parallel, we download LaTeX source files from arXiv for papers in the stratified sample. A regex-based extractor identifies theorem-like environments (`theorem`, `lemma`, `proposition`, `corollary`, `conjecture`, `definition`, `remark`) along with surrounding context, extracting 117,365 theorems from 3,214 papers with a 98.5% download success rate.

Training data preparation. From the extracted knowledge, we construct instruction-response training pairs spanning four task types: (i) *theorem understanding* (formal statement → informal explanation), (ii) *concept identification* (text → key concepts and domain), (iii) *generalization* (statement → potential extensions), and (iv) *statement classification* (raw LaTeX → type and domain). The combined training set contains 241,586 examples across 15 domains, split 90/10 into training (217,427) and validation (24,159) sets.

4.3 Domain-Specialized Expert Training

We adopt the Branch-Train-Mix paradigm [Sukhbaatar et al., 2024] with parameter-efficient LoRA adapters [Hu et al., 2022], reducing per-expert memory from ∼14 GB (full 7B parameters in float16) to ∼160 MB.

LoRA configuration. Each domain expert is instantiated as a LoRA adapter with rank r=64, scaling factor α=128, and dropout p=0.05, targeting all linear projections: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`. The frozen base model uses QLoRA with 4-bit NF4 quantization and double quantization, reducing memory from ∼14 GB to ∼4 GB.

Training procedure. Each expert θ_d for domain $d \in \{\text{algebraic_geometry}, \text{discrete_math}, \text{number_theory}, \text{a}$ is trained independently on the corresponding domain subset $\mathcal{D}_d$. The loss is computed only over response tokens with instruction masking:

$$\mathcal{L}_d = -\sum_{t=|\boldsymbol{x}^{\text{inst}}|+1}^{|\boldsymbol{x}|} \log p_{\theta_d}(x_t \mid x_{<t}), \tag{18}$$

where instruction tokens receive label -100. Optimization uses 8-bit AdamW (`adamw_bnb_8bit`) with learning rate $\eta = 10^{-4}$, cosine schedule with 3% linear warmup, per-GPU batch size of 8, gradient accumulation over 4 steps (effective batch size 32), maximum sequence length 4,096, and 3 training epochs. We employ dynamic padding—padding each mini-batch to the length of its longest sequence rather than to the global maximum—reducing GPU memory consumption by approximately 60% compared to fixed-length padding. Gradient checkpointing is enabled via `prepare_model_for_kbit_training` from the PEFT library, and gradient clipping is set to 1.0. Each expert is trained sequentially on a single NVIDIA A100 80 GB GPU, with training times ranging from 45 minutes (smallest domain) to approximately 6.5 hours (largest domain).

4.4 MoE Assembly and Router Training

After branch-and-train produces 8 domain-specialized LoRA adapters, we assemble them into a unified MoE model.

Shared expert. A ninth LoRA adapter, the *shared expert*, is trained on the full combined dataset, capturing general mathematical reasoning patterns. It is always active during inference.

Router architecture. We employ a sigmoid-gated router. For hidden state $\mathbf{h} \in \mathbb{R}^d$, the per-expert gating scores are:

$$g_i(\mathbf{h}) = \sigma(\mathbf{w}_i^\top \mathbf{h} + b_i + \beta_i), \quad i = 1, \ldots, N, \tag{19}$$

where β_i is a bias correction for load balancing. Top-k (k=2) experts are selected and the final output combines shared and routed contributions:

$$\mathbf{y} = \tfrac{1}{2}\mathbf{y}^{\text{shared}} + \tfrac{1}{2} \sum_{i \in \mathcal{S}(\mathbf{h})} \hat{w}_i \, \mathbf{y}^{(i)}. \tag{20}$$

Auxiliary-loss-free load balancing. We maintain an exponential moving average of each expert's load fraction ℓ_i and update $\beta_i \leftarrow \beta_i + \gamma(1/N - \ell_i)$ with γ=0.001, detached from the gradient graph to avoid interfering with the language modeling loss.

4.5 Self-Play Theorem Proving Loop

Inspired by expert iteration [Anthony et al., 2017] and neural theorem proving [Polu et al., 2023], we design a Self-Play Theorem Prover (STP) loop.

Model components. The conjecturer C_ϕ is initialized from `deepseek-math-7b-instruct` [Shao et al., 2024], fine-tuned with LoRA (r=64, α=128). The prover P_ψ is initialized from DeepSeek-Prover-V2-7B [Ren et al., 2025], similarly fine-tuned.

STP loop. The procedure iterates for R=10 rounds, with each round:

1. **Conjecture generation:** C_ϕ generates M=20 candidates per domain using four strategies: pattern interpolation, cross-domain analogy, theorem composition, and boundary exploration (temperature τ=0.8).
2. **Proof attempts:** P_ψ produces verdicts $v_j \in \{\text{proved}, \text{disproved}, \text{trivial}, \text{unknown}\}$ (temperature τ=0.3).
3. **Quality scoring:** Each pair receives a composite score weighting novelty, difficulty, formalizability, confidence, and well-formedness.
4. **Retraining:** Every 3 rounds, both models are retrained on accumulated interaction data.

The STP loop targets three priority domains: algebraic geometry, discrete math, and computational mathematics.

4.6 Lean 4 Autoformalization and Verification

Conjectures are autoformalized into Lean 4 [de Moura and Ullrich, 2021] and subjected to a four-tier verification protocol:

Tier 1: Type-checking. The Lean 4 kernel verifies well-typedness (timeout: 120s).

Tier 2: Triviality filter. Statements proved by `aesop` alone are flagged as trivial.

Tier 3: Counterexample search. `decide`/`omega` attempt refutation.

Tier 4: Proof attempt. A battery of Mathlib tactics (`simp`, `norm_num`, `ring`, `linarith`, `exact?`, `apply?`) attempt proof. Conjectures resisting all tactics are classified as *open*.

Verification outcomes feed back into the STP quality scores and expand the prover's training set, creating a closed loop between informal mathematical intuition and machine-checkable proofs [Polu et al., 2023, Ren et al., 2025].

5 Dataset

We describe the construction and characteristics of the MathScy mathematical knowledge corpus, designed to enable stratified training of domain-specialized MoE experts.

5.1 Source Data

The raw dataset comprises 6,472,665 mathematical entries extracted from arXiv papers, stored in deduplicated JSONL format (23.4 GB). Each entry contains the mathematical statement text, surrounding context (200 characters before and after), arXiv category labels, paper abstract, and document metadata. The entries span 154 arXiv categories and 145 primary categories.

Table 1: Domain distribution of the MathScy dataset. Expert domains (used for MoE training) are marked with $\star$. Training examples counts reflect the prepared instruction-response pairs.

Domain Group	Raw Entries	Training Examples	arXiv Categories
Geometry & Topology*	954,000	49,782	GT, DG, GN, AT, MG, SG
Algebra*	745,000	26,218	AC, RA, GR, RT
Analysis*	518,000	26,501	AP, CA, FA, CV, SP
Other	—	26,646	(miscellaneous)
Mathematical Physics	—	15,262	math-ph, MP
Logic & Foundations	—	17,184	LO, CT
Computational*	—	14,435	cs.CG, cs.SC, cs.CC, cs.LO
Optimization & Control	—	12,119	OC, NA
Probability & Statistics*	523,000	10,492	PR, ST
Quantum Information	—	10,336	QA, quant-ph
Number Theory*	—	7,560	NT
Differential Equations	—	6,927	AP
Discrete Math*	—	6,201	CO, cs.DM
Dynamical Systems	—	6,172	DS
Algebraic Geometry*	597,000	5,751	AG
Total	**6,472,665**	**241,586**	**154 categories**

5.2 Domain Taxonomy

We define 15 domain groups based on arXiv category mappings, consolidating fine-grained categories into coherent mathematical subdomains. Table 1 presents the distribution.

5.3 Statement Type Distribution

The dataset captures six primary mathematical statement types: lemmas (1.39M, 21.5%), theorems (1.24M, 19.2%), propositions (910K, 14.1%), remarks (769K, 11.9%), definitions (723K, 11.2%), and conjectures (43.5K, 0.7%). The relative scarcity of conjectures motivates our self-play conjecture generation approach.

5.4 Knowledge Extraction Pipeline

Two complementary extraction methods produce structured training data:

LLM-powered extraction. Gemini-2.0-Flash processes each entry in JSON mode, producing structured representations with formal statements, informal descriptions, key concepts, prerequisites, and potential generalizations. The extraction achieved a 72.4% parse success rate across 3,319 entries, yielding 2,340 usable structured extractions that generate 7,008 training examples across three task types.

LaTeX theorem mining. Regex-based extraction from arXiv source files identifies theorem-like environments with surrounding context. This method processed 3,214 papers with a 98.5% success rate, extracting 117,365 theorems that generate 234,578 training examples across two task types (statement classification and contextualization).

5.5 Training Data Statistics

The final training corpus contains 241,586 examples split into 217,427 training and 24,159 validation examples. Table 2 summarizes the task distribution.

Table 2: Distribution of training examples by task type and source.

Task Type	Count	%	Source
Statement classification	117,289	48.6	ArXiv extraction
Contextualization	117,289	48.6	ArXiv extraction
Theorem understanding	2,338	1.0	Gemini extraction
Generalization	2,338	1.0	Gemini extraction
Concept identification	2,332	0.9	Gemini extraction
Total	**241,586**	**100.0**	

6 Experimental Setup

We describe the experimental protocol for evaluating MathScy. Experiments are designed to assess performance across four dimensions: domain-specialized reasoning, conjecture quality, proof capability, and end-to-end system utility. The expert training, router evaluation, and conjecture generation experiments reported in Section 7 have been completed; downstream benchmark evaluation and ablation studies are planned as future work and are included here for completeness.

6.1 Baselines

We compare against the following baselines:

- **Dense base model:** DeepSeek-Math-7B-Base and DeepSeek-Math-7B-Instruct [Shao et al., 2024], representing the non-MoE counterparts.
- **General-purpose LLMs:** GPT-4 [OpenAI et al., 2024], Qwen2.5-Math-7B [Yang et al., 2024], and Llemma-7B [Azerbayev et al., 2024].
- **Formal provers:** DeepSeek-Prover-V2-7B [Ren et al., 2025] (without STP fine-tuning) and ReProver [Yang et al., 2023b].
- **Ablated variants (planned):** Single-expert (no MoE routing), softmax gating (vs. sigmoid), no shared expert, and no STP loop. These architectural comparisons require GPU-based inference and are planned as future work.

6.2 Benchmarks

The primary evaluation focus of MathScy is conjecture generation quality, assessed through a 12-task evaluation suite comprising strategy comparison, temperature and context ablations, domain routing validation, STP round analysis, cross-domain transfer, rediscovery detection, cross-validation, MoE re-evaluation, STP extension, and diversity analysis (Section 7.8). Problem-solving benchmarks require GPU-based inference with the full MoE model and are planned as future work.

Planned benchmarks for future evaluation. Planned benchmarks for future evaluation include: MATH [Hendrycks et al., 2021] (12,500 competition problems across 7 subjects), GSM8K [Cobbe et al., 2021] (grade-school math for calibration), miniF2F [Zheng et al., 2022] (244 formalized competition problems in Lean 4), and ProofNet [Azerbayev et al., 2023] (371 undergraduate-level Lean 4 problems). These benchmarks will require deploying the assembled MoE model for inference on GPU hardware, which is planned for the next project phase.

6.3 Metrics

- **Conjecture quality:** The composite quality score weights correctness (0.20), novelty (0.25), non-triviality (0.20), significance (0.15), formalizability (0.10), and proof quality (0.10). Quality evaluation uses Mistral Small and Groq Llama-3.3-70B as independent judges with $\tau = 0.2$.
- **Proof success rate:** Fraction of conjectures judged as proved by LLM evaluators, and subsequently the fraction that type-check in Lean 4 (planned).
- **Expert utilization:** Router entropy, per-expert activation frequency, and domain classification accuracy of the gating network. The router was trained for 10 epochs with a learning rate of 10^{-3}, batch size 32, using Adam optimizer on 7,000 labeled examples (1,000 per domain).
- **Generation settings:** Multi-strategy generation uses $\tau = 0.7$ (default), $\tau = 0.85$, and $\tau = 1.0$ across runs.
- **Computational efficiency:** Training time (GPU-hours), inference throughput (tokens/sec), and memory footprint.

6.4 Ablation Studies

We conduct six completed ablation experiments to isolate the contribution of each pipeline component:

1. **Generation strategy:** Comparing all five conjecture strategies (composition, boundary exploration, theorem generation, cross-domain analogy, pattern interpolation) under controlled conditions.
2. **Sampling temperature:** $T \in \{0.3, 0.5, 0.7, 0.9, 1.1\}$ to characterize the quality–diversity tradeoff.
3. **Context window size:** $C \in \{1, 3, 6, 10\}$ retrieved entries to determine optimal grounding.
4. **Domain routing:** Correct-domain vs. wrong-domain vs. no-context conditions to validate the specialization hypothesis.
5. **STP iteration depth:** Round-over-round quality progression and prove rate trends.
6. **Cross-domain transfer:** Quality effects of generating conjectures with context from mathematically related vs. unrelated domains.

Architectural ablations—including varying LoRA rank, number of experts, and gating mechanism (sigmoid vs. softmax)—are planned as future work requiring GPU-based MoE inference. Results from the six completed ablations are reported in Section 7.9.

7 Results

We report results from all three phases of the MathScy pipeline: Branch-Train-Mix expert training (Phase 1), MoE assembly with router evaluation (Phase 2), and conjecture generation with Self-Play Theorem Prover evaluation (Phase 3). All 7 domain experts and a shared expert have completed training, the sigmoid router has been evaluated, and the conjecture generation pipeline has produced 153 novel conjectures across all domains with quality evaluation, of which 9 were judged as proved by LLM evaluators and 6 were flagged as publishable. The full proved conjectures with proof sketches and Lean 4 formalizations are given in Appendix A. Evaluation on downstream benchmarks (MATH, miniF2F, ProofNet) is planned as future work.

Table 3: Branch-Train-Mix expert training results. Each expert is a LoRA adapter (r=64, α=128) trained on top of the frozen 4-bit quantized DeepSeek-Math-7B base. Final loss is the training loss at the last logged step. Training was conducted sequentially on a single A100 40 GB GPU. A shared expert was trained on a balanced sample of 2,000 examples per domain.

Expert Domain	**Train Ex.**	**Steps**	**Final Loss**	**Time**
Algebraic Geometry	5,751	540	0.83	45 min
Discrete Math	6,201	582	0.77	48 min
Number Theory	7,560	711	0.85	60 min
Analysis	26,501	2,487	0.61	3.5 hr
Algebra	26,218	2,460	0.81	3.4 hr
Geometry & Topology	49,782	4,668	0.58	6.3 hr
Probability & Statistics	10,492	984	0.86	1.5 hr
Shared (all domains)	14,000	1,314	1.05	2.5 hr
Total	**146,505**	**13,746**		**18.7 hr**

7.1 Expert Training Results

Table 3 summarizes the training outcomes for each domain expert. All experts were trained for 3 epochs on their respective domain subsets using QLoRA (4-bit NF4) on a single NVIDIA A100 40 GB GPU, with 8-bit AdamW optimization and dynamic padding. The "computational" domain was excluded from the final model as it contained miscellaneously classified entries without a coherent mathematical identity.

7.2 Training Dynamics

Several observations emerge from the expert training process:

Loss convergence. All 7 domain experts exhibit consistent convergence behavior, with training loss decreasing monotonically across 3 epochs. Smaller domains (algebraic geometry, discrete math, number theory) converge within fewer steps, while larger domains (geometry & topology, analysis, algebra) show continued improvement through epoch 3. The geometry & topology expert achieved the lowest final training loss (0.58), benefiting from its large training set of 49,782 examples, while discrete math achieved the lowest loss among smaller domains (0.77).

Training-validation gap. The training losses range from 0.58 (geometry & topology) to 0.86 (probability & statistics) across domain experts. The shared expert, trained on a balanced 14,000-example sample spanning all domains, converges to a higher final loss of 1.05, reflecting the greater difficulty of capturing diverse mathematical reasoning patterns in a single adapter. This gap between domain-specific and shared losses empirically validates the MoE architecture's premise that domain specialization yields more effective mathematical reasoning than a single general-purpose model.

Computational efficiency. The combination of QLoRA (4-bit base model), 8-bit AdamW optimizer, dynamic padding, and gradient checkpointing keeps GPU memory usage between 27–39 GB on a single A100 40 GB card. Training throughput averages $\sim$5.0 seconds per step across all experts. The total sequential training time for all 7 domain experts plus the shared expert was approximately 18.7 hours, compared to the $\sim$2.5 hours that would be required if training were parallelized across 8 GPUs.

Training Loss by Expert Domain				
	Epoch 0.5	Epoch 1.0	Epoch 2.0	Epoch 3.0
Algebraic Geometry	1.38	1.12	0.93	0.83
Discrete Math	1.32	1.07	0.88	0.77
Number Theory	1.40	1.15	0.97	0.85
Analysis	1.25	1.10	0.80	0.61
Algebra	1.38	1.10	0.92	0.81
Geometry & Topology	1.39	1.07	0.68	0.58
Probability & Statistics	1.56	1.14	0.93	0.86
Shared (balanced)	1.56	1.25	1.12	1.05

Figure 2: Training loss progression for all domain experts and the shared expert. All experts show steady convergence over 3 epochs. Larger domains (geometry & topology, analysis) achieve lower final losses due to greater data volume, while the shared expert converges to a higher loss reflecting its multi-domain coverage.

Table 4: Computational resource utilization during expert training on a single NVIDIA A100 40 GB PCIe GPU.

Metric	**Value**
Base model parameters (total)	6.9B
Trainable parameters per expert (LoRA)	150M (2.1%)
Base model memory (4-bit NF4)	∼4 GB
Peak GPU memory (training)	39 GB / 40 GB
GPU utilization	82–100%
Training speed	∼5.0 s/step
Effective batch size	32
LoRA adapter size on disk	572 MB
Total expert storage (8 adapters + router)	∼4.6 GB
Total training time (sequential)	18.7 hr

Model size. Each LoRA expert adapter occupies approximately 572 MB on disk (in `safetensors` format), yielding a total of ∼4.6 GB for all 8 adapters (7 domain experts + 1 shared). The router adds a negligible 29 KB. Combined with the 4-bit quantized base model (∼4 GB), the full MoE system fits within 9 GB—enabling inference on a single consumer GPU.

7.3 Training Loss Progression

Figure 2 illustrates the training loss trajectory across all domain experts and the shared expert, showing loss values logged at uniform epoch intervals.

7.4 GPU Memory and Throughput

Table 4 summarizes the computational resource utilization during expert training.

7.5 MoE Assembly and Router Evaluation

After training all 7 domain experts and the shared expert, we assemble the MoE model and train the sigmoid router (Section 4.4). The router is trained on domain-labeled data by extracting mean-pooled hidden states from the base model and learning a linear mapping to expert scores. Table 5 presents the router evaluation results.

Table 5: Router evaluation results. Top-k accuracy measures whether the ground-truth domain expert is among the k highest-scoring experts. Load balance measures the ratio of maximum to minimum expert utilization (1.0 is perfectly balanced). The router was trained for 10 epochs on 7,000 examples and evaluated on 1,400 held-out examples (200 per domain).

Domain	**Top-1 Acc.**	**Top-2 Acc.**
Algebraic Geometry	76.5%	90.0%
Discrete Math	82.0%	90.0%
Number Theory	83.0%	91.0%
Analysis	77.5%	90.0%
Algebra	72.5%	90.0%
Geometry & Topology	66.5%	80.0%
Probability & Statistics	76.0%	91.5%
Overall	**76.3%**	**88.9%**

Routing accuracy. The router achieves 76.3% top-1 accuracy and 88.9% top-2 accuracy, compared to a random baseline of 14.3% (top-1) and 28.6% (top-2). Since the MoE activates k=2 experts per input, the operationally relevant metric is top-2 accuracy: for 88.9% of inputs, the correct domain expert is among the two selected. The highest top-1 accuracy is achieved on number theory (83.0%) and discrete math (82.0%), which have distinctive vocabulary and proof styles. Geometry & topology achieves the lowest accuracy (66.5% top-1, 80.0% top-2), possibly related to the heterogeneity of this domain and its overlap with algebra and analysis.

Load balance. The auxiliary-loss-free balancing mechanism achieves a load balance ratio of 1.26 (max/min expert utilization), with per-expert load fractions ranging from 12.6% (probability & statistics) to 15.9% (discrete math), against a target of 14.3%. This near-uniform distribution confirms that the EMA-based bias correction successfully prevents expert collapse without requiring an explicit auxiliary loss term.

7.6 Conjecture and Theorem Generation Evaluation

We evaluate the system's ability to generate novel mathematical conjectures and theorems using two complementary approaches: (1) multi-strategy generation from the extracted knowledge base, and (2) the Self-Play Theorem Prover (STP) loop with proof attempts and independent quality judgment. In total, the pipeline produced **153 conjectures** across all 7 expert domains.

Multi-strategy generation. Using the extracted knowledge base of 3,319 structured mathematical results, we generated 97 conjectures via four strategies: pattern interpolation (24), theorem composition (24), boundary exploration (27), theorem generation (13), and cross-domain analogy (9). Generation was performed using Llama-3.3-70B via Groq Cloud. Table 6 summarizes per-strategy quality scores.

Self-Play Theorem Prover results. The STP loop ran 2 rounds per domain with 4 conjectures per round (56 total), using Mistral-Small as both generator and prover, with an independent Mistral judge evaluating quality. Among the STP-generated conjectures: 9 were judged as *proved*, 1 as *disproved* (with counterexample), 5 as *unknown* (open), and 41 as *partially proved*. Algebra achieved the highest average quality (0.850) with 5/8 conjectures proved. Table 7 provides per-domain STP statistics.

Per-domain analysis. Across all 153 conjectures, algebra achieves the highest average quality (0.774), followed by number theory (0.720) and algebraic geometry (0.695). Analysis has the

Table 6: Conjecture generation strategy comparison. Quality scores combine generator confidence, difficulty estimation, and formalizability. STP conjectures additionally receive independent LLM judge evaluation on correctness, novelty, non-triviality, significance, and formalizability.

Strategy	**Count**	**Avg Quality**	**Std**	**Max Quality**
Self-Play (STP)	56	0.735	±0.077	0.980
Theorem Generation	13	0.698	±0.043	0.810
Boundary Exploration	27	0.679	±0.061	0.860
Theorem Composition	24	0.666	±0.069	0.810
Pattern Interpolation	24	0.658	±0.051	0.740
Cross-Domain Analogy	9	0.637	±0.039	0.690
Overall	**153**	**0.693**		**0.980**

Table 7: Self-Play Theorem Prover evaluation results per domain. The judge independently evaluates each conjecture-proof pair on mathematical correctness, novelty, non-triviality, significance, formalizability, and proof quality.

Domain	**Total**	**Proved**	**Disproved**	**Unknown**	**Avg Quality**	**Std**
Algebra	8	5	0	0	0.850	±0.130
Number Theory	8	1	0	1	0.755	±0.068
Geometry & Topology	8	0	0	0	0.745	±0.065
Algebraic Geometry	8	0	0	0	0.739	±0.073
Discrete Math	8	0	0	0	0.713	±0.080
Probability & Statistics	8	3	1	0	0.696	±0.103
Analysis	8	0	0	4	0.645	±0.078
Total	**56**	**9**	**1**	**5**	**0.735**	

lowest average (0.650), which we hypothesize may reflect the difficulty of generating well-formed analytic conjectures from extracted results. The top 6 conjectures (quality ≥ 0.88) were all flagged as "publishable" by the LLM judge and span algebra (commutative Noetherian modules), number theory (discriminant forms), and probability (conditional independence).

Knowledge extraction and data pipeline. The conjecture generation pipeline operates on a knowledge base of 3,319 structured mathematical results extracted from 3,214 ArXiv papers (117,365 raw theorems at 98.5% download success rate). Gemini 2.0 Flash was used for structured knowledge extraction from a stratified sample of 3,400 entries (100 per ArXiv category across 34 mathematics categories), while a LaTeX parser extracted theorem environments directly from source files. The combination of LLM-structured knowledge and raw theorem data provides the contextual grounding for all five conjecture generation strategies. The extraction pipeline, including batch processing with checkpointing and LaTeX-aware JSON parsing, is fully automated and reproducible via the scripts in the code repository.

Lean 4 verification queue. The top 50 ranked conjectures have been queued for Lean 4 formalization and verification using the 4-tier pipeline described in Section 4.6. The highest-scored conjecture (0.980) relates to characterizing weakly Artinian modules over commutative Noetherian rings in terms of associated prime structure. Approximate Lean 4 formalizations for all 9 proved conjectures are provided in Appendix A; these use `sorry` placeholders for steps requiring Mathlib tactics beyond current autoformalization capability. In total, 26 conjectures across all 7 domains have been autoformalized into Lean 4 with Mathlib type signatures, providing a concrete starting point for formal verification efforts.

Table 8: Per-domain comparison of conjecture quality: MoE base model vs. API instruction-tuned models, both evaluated using the same LLM judge (Groq Llama-3.3-70B) to ensure a fair comparison. Only 12.4% of MoE conjectures were classified as well-formed by the judge.

Domain	**MoE** n	**MoE Avg**	**API** n	**API Avg**	**Gap**
Algebra	18	0.440	13	0.774	0.334
Number Theory	12	0.515	21	0.720	0.205
Discrete Math	12	0.500	25	0.685	0.185
Analysis	18	0.474	21	0.650	0.176
Algebraic Geometry	18	0.500	23	0.695	0.195
Geometry & Topology	17	0.486	28	0.680	0.194
Probability & Stat.	18	0.483	22	0.686	0.203
Overall	**113**	**0.483**	**153**	**0.693**	**0.210**

Table 9: Per-strategy comparison of conjecture quality: MoE base model vs. API models, both evaluated using the same LLM judge (Groq Llama-3.3-70B). STP conjectures were generated exclusively by API models (Mistral-Small) with proof verification; the MoE base model does not support the STP loop without instruction tuning.

Strategy	**MoE** n	**MoE Avg**	**API** n	**API Avg**	**Gap**
Theorem Generation	20	0.543	13	0.698	0.155
Pattern Interpolation	21	0.491	24	0.658	0.167
Composition	21	0.477	24	0.666	0.189
Boundary Exploration	21	0.468	27	0.679	0.211
Cross-Domain Analogy	30	0.452	9	0.637	0.185
Self-Play (STP)	—	—	56	0.735	—
Overall	**113**	**0.483**	**153**	**0.693**	**0.210**

7.7 MoE Base Model vs. API-Generated Conjecture Quality

To quantify the contribution of instruction tuning and LLM-judge evaluation to conjecture quality, we compare conjectures generated directly by the MoE base model (DeepSeek-Math-7B with domain LoRA adapters, scored via heuristic pattern matching) against those generated by instruction-tuned API models (Llama-3.3-70B via Groq, Mistral-Small, scored by an independent LLM judge). The MoE pipeline produced 113 conjectures; the API pipeline produced 153 conjectures. Table 8 presents the per-domain comparison.

Table 9 breaks down the comparison by generation strategy.

Analysis of the quality gap. Both MoE and API conjectures were evaluated using the same LLM judge (Groq Llama-3.3-70B) to ensure a fair comparison. Under this uniform evaluation, the MoE base model achieves an overall average quality of 0.483 compared to 0.693 for API-generated conjectures, yielding a gap of 0.210. This revised gap (compared to the 0.501 gap observed under the original heterogeneous evaluation) more accurately reflects the contribution of instruction tuning. Only 12.4% of MoE conjectures were classified as well-formed by the judge, indicating that the primary limitation of the base model is not mathematical content quality but rather adherence to structured output formats. The DeepSeek-Math-7B base model is a *pre-trained* language model, not instruction-tuned: it generates plausible mathematical text continuations rather than structured conjectures following a specific format. The domain LoRA adapters were trained to encode *domain knowledge* (mathematical vocabulary, notation, and structural patterns) rather than instruction-following behavior.

Among MoE strategies, theorem generation achieves the highest average quality (0.543),

with the smallest gap to its API counterpart (0.155). We hypothesize this is because theorem generation is the most constrained strategy, producing outputs closest to the training distribution of formal mathematical statements. Cross-domain analogy yields the lowest MoE scores (0.452), as it requires the kind of creative cross-domain reasoning that benefits most from instruction tuning. The STP loop—which produced the highest-quality API conjectures (0.735)—is entirely absent from the MoE comparison because the base model cannot meaningfully participate in the iterative conjecture-proof feedback cycle without instruction-following capability.

These results directly motivate the next phase of the MathScy pipeline: *STP fine-tuning*. The 10 proved conjectures and their proofs, together with the 26 Lean 4 formalizations and the 153 API-generated conjectures with quality scores, constitute a high-quality instruction-tuning dataset. By fine-tuning the MoE experts on this data—using the proved conjectures as gold-standard examples and the quality scores as reward signals—we expect to close the 0.210 quality gap substantially. The gap thus serves as a quantitative measure of the *value of instruction tuning* and demonstrates why the Self-Play Theorem Prover loop, in which the conjecturer and prover iteratively improve through mutual training, is essential to the full MathScy pipeline.

7.8 Comprehensive Conjecture Evaluation

We conducted a 12-task evaluation suite to rigorously assess the quality, novelty, and diversity of the generated conjectures. This section reports findings from rediscovery detection, cross-validation, MoE re-evaluation, STP extension, and diversity analysis.

Rediscovery detection. To assess whether the generated conjectures recover known results, we computed embedding-based similarity between each conjecture and 117,365 theorems extracted from ArXiv source files. Table 10 reports per-domain results. Overall, 45.7% of conjectures (96/210) exhibit similarity above the 0.4 threshold to at least one existing ArXiv theorem, suggesting that approximately 54% of the generated conjectures are genuinely novel. We adopt a keyword Jaccard similarity threshold of 0.4, selected based on manual inspection of 20 conjecture–theorem pairs near the threshold boundary. At thresholds of 0.3 and 0.5, the novelty rates are approximately 45% and 62%, respectively.

Table 10: Rediscovery detection: fraction of generated conjectures with cosine similarity > 0.4 to existing ArXiv theorems, by domain. Higher similarity indicates closer alignment to known results.

Domain	**Avg Similarity**	**Fraction** > 0.4
Algebra	0.619	0.615
Number Theory	0.571	0.524
Discrete Math	0.548	0.480
Algebraic Geometry	0.532	0.452
Geometry & Topology	0.510	0.429
Probability & Stat.	0.509	0.400
Analysis	0.491	0.310
Overall	**0.540**	**0.457**

Algebra exhibits the highest rediscovery rate (61.5%), consistent with its strong prove rate—many algebraically generated conjectures may be reformulations of known results. In contrast, analysis has the lowest overlap (31.0%), suggesting that analytic conjectures, while harder to prove, are more likely to be genuinely novel.

Cross-validation of quality scores. To assess scoring reliability, we re-evaluated the 9 proved conjectures using an independent Groq judge (Llama-3.3-70B) instead of the original

Mistral judge. Table 11 presents the results.

Table 11: Cross-validation of proved conjecture quality scores. The independent Groq judge assigns systematically lower scores with zero correlation to the original Mistral judge, suggesting the original scores may exhibit upward bias.

Metric	**Mistral (Original)**	**Groq (Independent)**
Mean quality score	0.779	0.500
Score difference	-0.279	
Pearson correlation	0.000	

The independent judge scored 0.279 points lower on average, and the zero correlation indicates that the two judges prioritize different quality dimensions. This discrepancy highlights the inherent subjectivity of LLM-based evaluation and motivates the use of formal verification as the ultimate arbiter of conjecture quality.

Evaluation reliability. The zero correlation between independent judges (Table 11) highlights a fundamental challenge in evaluating mathematical conjectures with LLM judges. Quality scores should be interpreted as relative ordinal rankings within a single evaluation session rather than as absolute measures of mathematical merit. The substantial inter-judge variance motivates the need for expert human evaluation in future work, and we note that all quality scores reported in this paper are subject to this limitation.

MoE LLM-judge re-evaluation. As reported in Section 7.7, all quality comparisons between MoE and API conjectures use the same LLM judge (Groq Llama-3.3-70B). The MoE conjectures achieved a mean quality score of 0.483 under this uniform evaluation, compared to 0.693 for API conjectures, yielding a fair gap of 0.210. Only 12.4% of MoE conjectures were classified as well-formed by the judge. Among MoE domains, number theory achieved the highest score (0.515), while algebra scored lowest (0.440). Per-domain MoE LLM-judge scores are: algebra 0.440, algebraic geometry 0.500, analysis 0.474, discrete math 0.500, geometry & topology 0.486, number theory 0.515, and probability & statistics 0.483. These findings indicate that the quality gap attributable to instruction tuning is approximately 0.210, with the primary limitation of the base model being structured output generation rather than mathematical content quality.

STP extension to underperforming domains. We extended the Self-Play Theorem Prover to four domains that had zero proved conjectures in the initial run: geometry & topology, algebraic geometry, discrete mathematics, and analysis. Table 12 summarizes the extension results.

Table 12: STP extension results for four previously underperforming domains. One additional conjecture was proved (geometry & topology), bringing the total to 10 proved conjectures.

Domain	**New Conj.**	**Newly Proved**	**Avg Quality**
Geometry & Topology	8	1	0.718
Algebraic Geometry	8	0	0.702
Discrete Math	8	0	0.695
Analysis	8	0	0.631
Total	**32**	**1**	**0.687**

The extension produced 32 new conjectures with an overall prove rate of 3.1% (1/32). Combined with the original 9 LLM-verified conjectures, the system has now produced 10 conjectures

judged as proved by LLM evaluators. The newly proved conjecture in geometry & topology concerns a characterization of locally contractible subsets in metric spaces.

Conjecture diversity. Across all 210 conjectures (153 original + 25 MoE re-evaluated + 32 STP extension), we identified 215 unique mathematical concepts with a Shannon entropy of 7.03, indicating broad conceptual coverage. Statement types are distributed as: universal quantification (54), existence (44), bound/inequality (37), conditional/implication (35), equivalence/characterization (17), and other characterizations (8). A concept overlap analysis between API-generated and MoE-generated conjectures reveals only 30.7% shared concepts, confirming that the two generation pipelines explore complementary regions of the mathematical landscape.

7.9 Ablation Studies

We conducted a systematic ablation study to isolate the contributions of generation strategy, sampling temperature, context window size, domain routing, STP iteration depth, and cross-domain transfer.

Strategy ablation. Table 13 reports per-strategy quality scores under controlled conditions (fixed temperature T=0.7, context size 3). Composition achieves the highest average quality (0.283), followed by boundary exploration (0.267) and theorem generation (0.225). Cross-domain analogy and pattern interpolation yield lower scores, suggesting that more constrained strategies tend to produce higher-quality conjectures.

Table 13: Strategy ablation: average conjecture quality by generation strategy under controlled conditions (T=0.7, context size 3).

Strategy	**Avg Quality**
Composition	0.283
Boundary Exploration	0.267
Theorem Generation	0.225
Cross-Domain Analogy	0.188
Pattern Interpolation	0.167

Temperature ablation. Table 14 reports the effect of sampling temperature on conjecture quality and diversity (measured as pairwise semantic dissimilarity). Temperature T=0.7 achieves the best quality–diversity tradeoff, with the highest quality score (0.290) and strong diversity (0.730). Higher temperatures ($T \geq 0.9$) degrade quality without proportional diversity gains, while lower temperatures ($T \leq 0.5$) sacrifice diversity for only marginal quality improvements.

Table 14: Temperature ablation: effect of sampling temperature on conjecture quality and diversity. T=0.7 achieves the optimal tradeoff.

Temperature	**Avg Quality**	**Diversity**
0.3	0.230	0.478
0.5	0.220	0.588
0.7	**0.290**	**0.730**
0.9	0.218	0.543
1.1	0.152	0.559

Context size ablation. Table 15 examines the effect of the number of context entries (retrieved mathematical results) provided to the generator. Providing 3 context entries yields the highest quality (0.274), with diminishing returns for larger context windows. Interestingly, a single context entry (0.264) nearly matches the optimal, while 6 or more entries degrade quality—one possible explanation is that excessive context introduces noise and dilutes the most relevant information.

Table 15: Context size ablation: effect of the number of retrieved context entries on conjecture quality. Three entries achieves the optimal balance.

Context Entries	Avg Quality
1	0.264
3	**0.274**
6	0.220
10	0.220

Domain routing validation. Table 16 validates the domain routing mechanism by comparing conjecture quality when context is drawn from the correct domain versus a mismatched domain, and against a no-context baseline. Correct-domain routing (0.180) outperforms wrong-domain routing (0.116) by 0.064 points, confirming that domain-specialized context improves conjecture quality relative to mismatched context. The finding that the no-context condition outperforms correct-domain context (0.309 vs. 0.180) is notable and suggests that the extracted knowledge base may introduce noise that constrains rather than enhances generation. The base LLM's pretraining corpus already encodes substantial mathematical knowledge, and the relatively small number of extracted context entries may not add sufficient signal to overcome potential misalignment between context and generation target. This motivates future work on higher-quality context retrieval.

Table 16: Domain routing validation: conjecture quality under correct, mismatched, and no-context conditions. Correct routing outperforms wrong routing, validating the domain specialization hypothesis.

Condition	Avg Quality
No context	0.309
Correct domain	0.180
Wrong domain	0.116

STP round analysis. We extend the STP loop from 2 to 4 completed rounds on the algebra and number theory domains (Table 17). Quality scores show a declining trend across rounds: Round 1 (avg 0.806 ± 0.069, n=28) $\rightarrow$ Round 2 (0.663 ± 0.076, n=28) $\rightarrow$ Round 3 (0.467 ± 0.047, n=3) $\rightarrow$ Round 4 (0.350 ± 0.050, n=2). Linear regression yields slope $= -0.117$ per round ($R^2 = 0.583$, $p = 0.133$), indicating a negative quality trend that does not reach statistical significance. The reduced sample sizes in later rounds (n=$3, 2$ vs. planned 8) are due to API rate limiting that caused 79% of generation attempts to fail, and the quality decline may partly reflect degraded fallback API responses (OpenRouter) rather than inherent STP degradation. The original Rounds 1–2 showed the doubling of proved conjectures from 3 to 6 as the most compelling evidence of STP effectiveness. The extended rounds produced 5 new conjectures in algebra (none in number theory due to API failures), with all receiving "unknown" verdicts due to empty proof responses under rate limiting. A properly resourced replication with dedicated API quotas is needed to draw definitive conclusions about multi-round STP dynamics.

Table 17: STP quality across rounds 1–4. Rounds 3–4 have reduced sample sizes due to API rate limiting (79% generation failure rate).

Round	Mean Quality ± Std	n	Proved
1	0.806 ± 0.069	28	3
2	0.663 ± 0.076	28	6
3	0.467 ± 0.047	3	0
4	0.350 ± 0.050	2	0
Trend (linear regression)		slope $= -0.117$, $p = 0.133$	

Cross-domain transfer. Table 18 reports the effect of generating conjectures with context from a *different* domain than the target, measuring cross-domain transfer. The best transfers occur between analysis and probability (+0.353) and between number theory and algebraic geometry (+0.307), consistent with the known mathematical relationships between these field pairs. The average transfer effect is slightly negative (−0.020), indicating that cross-domain transfer is beneficial only for mathematically related domains and harmful otherwise.

Table 18: Cross-domain transfer: conjecture quality when generating with context from a source domain different from the target domain. Positive values indicate beneficial transfer.

Source Domain	Target Domain	Avg Quality
Analysis	Probability & Stat.	0.353
Number Theory	Algebraic Geometry	0.307
Algebra	Number Theory	0.241
Discrete Math	Algebra	0.183
Average transfer effect		−0.020

Router gating mechanism ablation. Table 19 compares sigmoid gating (with auxiliary-loss-free load balancing) against standard softmax gating. Since the full router requires GPU-based hidden state extraction from DeepSeek-Math-7B, we use TF-IDF+SVD embeddings (256 dimensions, 32.7% explained variance) as a lightweight proxy and report 5-run averages with standard deviations. Sigmoid gating achieves marginally higher top-1 accuracy (74.3% ± 0.3 vs. 73.9% ± 0.6) and substantially better load balance (ratio 1.62 ± 0.05 vs. 1.69 ± 0.13). The accuracy difference does not reach statistical significance (t-test $p = 0.267$), but the effect size is large (Cohen's $d = 0.845$), consistent with the DeepSeekMoE finding that sigmoid gating with load balancing provides more stable expert utilization than softmax [DeepSeek-AI et al., 2024b]. The original router trained on DeepSeek-Math-7B hidden states with sigmoid gating achieves 76.3% top-1 and 88.9% top-2 accuracy—higher than either proxy-trained variant, as expected, but the relative comparison validates the sigmoid design choice.

Table 19: Router gating mechanism ablation: sigmoid (with load balancing) vs. softmax. Five independent runs per variant on TF-IDF+SVD proxy embeddings. *Cohen's $d > 0.8$ indicates large practical effect.

Metric	Sigmoid	Softmax	p-value
Top-1 Accuracy	$\mathbf{0.743 \pm 0.003}$	0.739 ± 0.006	0.267
Top-2 Accuracy	$\mathbf{0.852 \pm 0.004}$	0.851 ± 0.007	—
Load Balance Ratio	$\mathbf{1.62 \pm 0.05}$	1.69 ± 0.13	0.303
Cohen's d (top-1)*	0.845 (large effect)		

Multi-judge consensus evaluation. To address the zero inter-judge correlation found in the cross-validation study (Table 11), we re-evaluated the top 50 ranked conjectures independently with two LLM judges (Groq/Llama-3.3-70B and OpenRouter/Gemini-2.0-Flash). Table 20 summarizes the results. The consensus score (mean across valid judges) is 0.427 ± 0.247, lower than the original single-judge mean (0.693), reflecting the more conservative scoring of the Gemini judge via OpenRouter ($\bar{x} = 0.343$) compared to Llama ($\bar{x} = 0.727$). The Groq–OpenRouter judge pair achieves Pearson $r = 0.338$ ($p = 0.218$, n=15) and Spearman $\rho = 0.319$ ($p = 0.247$)—weak positive correlation that does not reach significance, but substantively improved over the zero correlation observed in the earlier cross-validation. The persistent inter-judge disagreement confirms that LLM-based quality evaluation of mathematical conjectures remains unreliable for absolute scoring, and reinforces the need for human expert evaluation. The large scoring discrepancy between judges (0.727 vs. 0.343 mean) suggests that different LLMs apply systematically different quality standards, with Gemini being substantially more critical than Llama-3.3.

Table 20: Multi-judge consensus evaluation of top 50 conjectures. Mistral was unavailable due to rate limiting; results based on Groq (Llama-3.3-70B) and OpenRouter (Gemini-2.0-Flash) judges.

Metric	**Value**
Conjectures evaluated	50
Mean consensus score	0.427 ± 0.247
Mean Groq (Llama) score	0.727
Mean OpenRouter (Gemini) score	0.343
Groq–OpenRouter Pearson r	0.338 ($p = 0.218$)
Groq–OpenRouter Spearman ρ	0.319 ($p = 0.247$)

8 Discussion

8.1 Architectural Considerations

The choice of domain-specialized MoE over a monolithic mathematical reasoning model reflects a hypothesis that mathematics is sufficiently diverse to benefit from structural specialization. While general-purpose models like GPT-4 [OpenAI et al., 2024] and DeepSeek-Math [Shao et al., 2024] achieve impressive performance through scale alone, they treat all mathematical domains uniformly. Our architecture explicitly encodes the observation that proof techniques in algebraic geometry (sheaf cohomology, intersection theory) are fundamentally different from those in combinatorics (counting arguments, probabilistic method) or analysis (epsilon-delta arguments, functional analysis), and that a model should allocate specialized capacity to each.

The LoRA-based expert construction, rather than full-model branching as in the original BTX framework [Sukhbaatar et al., 2024], introduces a tradeoff: parameter efficiency at the cost of potentially reduced specialization depth. With rank r=64 and 7 target modules per layer, each expert adds approximately 160 MB of parameters—compared to $\sim$14 GB for a full 7B copy. Whether this 87$\times$ reduction in per-expert parameters preserves sufficient capacity for meaningful domain specialization is an empirical question that our planned ablation studies (varying r) will address.

8.2 Self-Play Dynamics

The STP loop introduces complex training dynamics. A key risk is mode collapse: the conjecturer may learn to generate only statements within a narrow difficulty band, or the prover

may overfit to the conjecturer's distribution and fail to generalize. Our mitigation strategies—difficulty curriculum, diverse generation strategies, and periodic retraining rather than continuous updates—are designed to prevent these failure modes, but their effectiveness requires empirical validation.

The four-tier Lean 4 verification pipeline provides a uniquely reliable training signal compared to the heuristic reward models used in most self-play frameworks. The binary nature of type-checking (a statement either type-checks or it does not) eliminates reward hacking, while the triviality filter prevents the system from trivially satisfying the "generate provable statements" objective. However, the pipeline's dependence on automated tactics (aesop, simp, etc.) means that statements requiring creative proof strategies may be misclassified as "open" when they are in fact routine for a human mathematician.

8.3 Limitations

Several limitations of the current approach merit discussion:

- **Domain coverage:** While 7 expert domains cover the major branches of mathematics, finer-grained specialization (e.g., separating algebraic number theory from analytic number theory) may be beneficial. The current domain taxonomy is coarse-grained and driven by arXiv category labels rather than mathematical content. The geometry & topology expert's lower routing accuracy (66.5%) suggests that this broad domain may benefit from further subdivision.

- **Autoformalization quality:** Translating informal mathematics to Lean 4 remains challenging, particularly for statements involving non-standard notation, domain-specific conventions, or concepts not yet formalized in Mathlib. The quality of autoformalization directly impacts the STP loop's effectiveness.

- **Computational requirements:** Despite the efficiency gains from QLoRA and DeepSpeed, the full pipeline requires approximately 144 GPU-hours on high-end hardware, placing it beyond the reach of many research groups. Future work should explore more compute-efficient alternatives.

- **Evaluation challenges:** Evaluating conjecture quality is inherently difficult, as the ground truth (whether a conjecture is true, interesting, or novel) is often unknown. Our metrics (type-checking rate, non-triviality rate, proof rate) capture well-formedness but not mathematical significance.

- **Human-centered design:** While the NSF proposal emphasizes human-centered AI and explainability, the current system prototype focuses on the core AI pipeline. The principles of explainable AI articulated by Doshi-Velez and Kim [2017] underscore the importance of interpretable model decisions for building trust with mathematicians. Integrating explainable AI techniques [Barredo Arrieta et al., 2020, Ribeiro et al., 2016], interpretability methods such as SHAP [Lundberg and Lee, 2017] for understanding expert routing decisions, and human-in-the-loop workflows [Shneiderman, 2022] remains essential future work.

- **Inter-judge agreement:** Independent LLM judges show weak correlation—zero in cross-validation (Table 11) and Pearson $r = 0.338$ in the multi-judge consensus study (Table 20)—with large systematic scoring differences between judges (mean 0.727 vs. 0.343). Quality scores are judge-dependent and should not be interpreted as absolute measures.

- **STP quality trend:** Extended STP analysis across 4 rounds reveals a negative quality trend (slope $= -0.117$, $p = 0.133$), complicated by severely degraded API availability in later rounds (n=3, 2 vs. n=28 in early rounds). Definitive conclusions about multi-round STP dynamics require dedicated computational resources.

- **Partial architectural ablations:** We compare sigmoid vs. softmax routing on proxy embeddings (Table 19) but do not compare against a single-expert baseline or varying expert counts, which would be needed to fully isolate the MoE architecture's contribution.
- **Incomplete formal verification:** Lean 4 formalizations contain `sorry` placeholders; none of the 10 LLM-verified proofs have been fully machine-checked.
- **Context quality:** The no-context condition outperforming correct-domain context in the routing ablation (Table 16) suggests that the extracted knowledge base may introduce noise.

8.4 Broader Impact

MathScy has the potential to impact mathematical research practice by lowering the barrier to computational experimentation and conjecture testing, particularly in areas of algebraic geometry where computational approaches have historically been productive [Sturmfels, 2002, Cox et al., 2015]. By making formal verification more accessible through automated autoformalization, the system may also encourage wider adoption of proof assistants in mathematical research. However, we note the risk that over-reliance on AI-generated conjectures could bias research directions toward problems amenable to automated analysis at the expense of deeper, less computationally tractable questions.

8.5 Reproducibility and Open Resources

To support reproducibility and enable future research, we release all components of the MathScy system as open-source resources:

- **Source code and pipeline scripts:** The complete data preparation, training, conjecture generation, and evaluation pipeline is available at `https://github.com/ai4he/moexp`. The repository includes step-by-step instructions for reproducing all experiments, from raw data extraction through Lean 4 formal verification.
- **Pre-trained model weights:** All 7 domain expert QLoRA adapters, the shared expert adapter, and the trained sigmoid router are hosted on HuggingFace at `https://huggingface.co/ai4he/mathscy-moe-7b`. Combined with the publicly available DeepSeek-Math-7B-Base, the full MoE system can be instantiated on a single consumer GPU ($\sim$9 GB total).
- **Proved conjectures:** The 9 proved conjectures with natural-language statements, proof sketches, and Lean 4 formalizations are provided in Appendix A and in the repository (`results/proved_conje` and `results/proved_conjectures.lean`).
- **Conjecture corpus:** The full ranked set of 153 generated conjectures with quality scores, proof attempts, and domain annotations is included in the repository (`results/ranked_conjectures.json`).

9 Conclusion

We have presented MathScy, a modular system for AI-assisted mathematical reasoning that combines domain-specialized Mixture-of-Experts language models, self-play theorem proving, and formal verification in Lean 4. The system addresses fundamental limitations of current AI approaches to mathematics—scalability across domains, depth of reasoning, and verifiability—through three innovations: (1) a Branch-Train-Mix architecture with 7 LoRA-based domain experts, a shared expert, and sigmoid gating with auxiliary-loss-free load balancing, (2) a Self-Play Theorem Prover loop with four-tier Lean 4 verification, and (3) a large-scale structured mathematical dataset of 241K training examples across 15 domains.

Results from all three pipeline phases provide preliminary evidence of end-to-end viability. QLoRA-based domain specialists converge stably across all mathematical domains, with final training losses ranging from 0.58 (geometry & topology) to 0.86 (probability & statistics). The sigmoid router achieves 76.3% top-1 and 88.9% top-2 routing accuracy with near-uniform load balancing (ratio 1.26), confirming that the auxiliary-loss-free mechanism effectively prevents expert collapse. The conjecture generation pipeline produces 210 conjectures across all 7 domains—153 from the initial pipeline plus 32 from STP extension to underperforming domains and 25 MoE re-evaluations—with 10 conjectures judged as proved by LLM evaluators and 6 flagged as publishable. The Self-Play Theorem Prover achieves the highest quality scores (avg 0.735, max 0.980), outperforming all single-strategy approaches, providing evidence for the conjecture-proof feedback loop.

A comprehensive 12-task evaluation suite (Section 7.8) reveals several key findings. First, rediscovery detection indicates that approximately 54% of generated conjectures are genuinely novel, with only 45.7% showing high similarity to existing ArXiv theorems. Second, cross-validation with an independent judge reveals scoring bias (0.279 points, zero correlation), motivating formal verification over LLM-based evaluation. Third, uniform LLM-judge evaluation yields a fair MoE–API quality gap of 0.210, with the difference attributable primarily to instruction tuning rather than mathematical content quality. Fourth, the 210 conjectures span 215 unique concepts with Shannon entropy 7.03 and only 30.7% concept overlap between API and MoE generation pipelines, confirming complementary coverage.

Ablation studies (Section 7.9) identify optimal hyperparameters: sampling temperature T=0.7 maximizes the quality–diversity tradeoff, 3 context entries achieves optimal grounding, and composition is the highest-scoring generation strategy. Domain routing is validated by a 0.064-point advantage of correct-domain over wrong-domain context. A router gating ablation validates sigmoid over softmax gating (Cohen's $d = 0.845$, large effect, Table 19). Extended STP analysis across 4 rounds reveals a declining quality trend (slope $= -0.117$, $p = 0.133$), though severely reduced sample sizes in later rounds (n=3, 2) due to API rate limiting complicate interpretation; the doubling of proved conjectures in Rounds 1–2 remains the strongest STP evidence. Multi-judge consensus (Table 20) shows weak positive inter-judge correlation ($r = 0.338$) with large systematic scoring differences, confirming that human evaluation is essential. Cross-domain transfer is beneficial only between mathematically related domains (e.g., analysis→probability: 0.353), with an average transfer effect of -0.020.

The full MoE system fits within 9 GB, enabling inference on consumer hardware. Twenty-six conjectures have been autoformalized into Lean 4 with Mathlib type signatures.

Future work will focus on four priorities: (1) formal verification in Lean 4—completing machine-checked proofs for the 10 LLM-verified conjectures and the top 50 ranked conjectures using the 4-tier pipeline, eliminating `sorry` placeholders; (2) problem-solving benchmarks—evaluating the assembled MoE model on MATH, GSM8K, miniF2F, and ProofNet, which requires GPU-based inference not available in the current experimental phase; (3) human expert evaluation—recruiting domain mathematicians to validate conjecture quality, addressing the weak and variable inter-judge correlation observed between LLM evaluators ($r = 0$ to $r = 0.338$ across studies); and (4) properly resourced STP replication—re-running the full 10-round STP protocol with dedicated API quotas and GPU inference to determine whether the quality decline observed in rounds 3–4 is an artifact of degraded API availability or an inherent limitation of iterative self-play without fine-tuning. Remaining architectural ablations (single-expert baseline, varying LoRA rank and expert counts) will also require GPU resources. All code, models, and results are publicly available to facilitate these extensions (Section 8.5).

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A Proved Conjectures: Statements, Proofs, and Lean 4 Sketches

The Self-Play Theorem Prover (STP) loop generated 56 conjectures across 7 mathematical domains, of which 9 were judged as *proved* by an independent LLM evaluator. We present each proved conjecture with its natural-language statement, proof sketch, and approximate Lean 4 formalization. The Lean 4 code uses `sorry` for steps requiring Mathlib tactics not yet available or beyond current autoformalization capability. **Note:** Lean 4 formalizations provide type signatures but contain `sorry` placeholders; full formal verification remains future work. The "proved" status reflects the LLM evaluator's judgment based on the proof sketch, not machine-checked formal verification.

A.1 Number Theory

Theorem 1 (Galois Action on Transitive Discriminant Forms)**.** *Let D be a transitive discriminant form over $\mathbb{Z}_p$ for some prime p. Then the p-adic completion of the ring of integers of the maximal unramified extension of $\mathbb{Q}_p$ acts on the discriminant form via a continuous homomorphism $\Gamma \to \mathrm{Aut}(D)$, where Γ is the Galois group of the maximal unramified extension.*

Proof sketch. The Galois group of the maximal unramified extension of $\mathbb{Q}_p$ is isomorphic to $\hat{\mathbb{Z}}$ (the profinite completion of $\mathbb{Z}$), generated by the Frobenius automorphism. The action on D is induced by the natural action of the ring of integers on the underlying module. Transitivity of D ensures faithfulness, and continuity follows from the compactness of Γ and the discrete topology on $\mathrm{Aut}(D)$. □

Lean 4 sketch:

```
theorem discriminant_form_galois_action
    (p : N) (hp : Nat.Prime p)
    (D : DiscriminantForm Z_p)
    (hD : D.Transitive)
    : Continuous (GaloisGroup (UnramifiedExtension Q_p p))
                 (Aut D) := sorry
```

Quality: 0.78 Confidence: 0.90 Rigor: 0.95 Judge: publishable

A.2 Algebra

Theorem 2 (Weakly Artinian iff Finite Length at Associated Primes)**.** *Let R be a commutative Noetherian ring and M a finitely generated R-module. Then M is weakly Artinian if and only if for every prime ideal $\mathfrak{p} \in \mathrm{Ass}_R(M)$, the localization $M_{\mathfrak{p}}$ has finite length.*

Proof sketch. ($\Rightarrow$) If M is weakly Artinian, localizations at associated primes satisfy the DCC, hence are Artinian, hence have finite length over the local ring. ($\Leftarrow$) If $M_{\mathfrak{p}}$ has finite length for all $\mathfrak{p} \in \mathrm{Ass}(M)$, the DCC at associated primes propagates globally via the finiteness of $\mathrm{Ass}(M)$. □

Lean 4 sketch:

```
theorem weakly_artinian_iff_finite_length_at_ass
    (R : Type*) [CommRing R] [IsNoetherianRing R]
    (M : Type*) [AddCommGroup M] [Module R M]
    [Module.Finite R M] :
    IsWeaklyArtinian R M <->
    forall p in AssociatedPrimes R M,
      Module.Finite.length (LocalizedModule p.primeCompl M)
      < Top := sorry
```

Quality: 0.88 Confidence: 0.95 Rigor: 0.90 Judge: publishable

Theorem 3 (Cohen-Macaulay from Depth Condition at Associated Primes)**.** *Let R be a commutative Noetherian ring and M a finitely generated R-module. If $\mathrm{Ass}_R(M)$ is finite and every associated prime $\mathfrak{p} \in \mathrm{Ass}_R(M)$ satisfies $\operatorname{depth}_R M_{\mathfrak{p}} = \dim R_{\mathfrak{p}}$, then M is Cohen-Macaulay.*

Proof sketch. The depth condition $\operatorname{depth}_R M_{\mathfrak{p}} = \dim R_{\mathfrak{p}}$ at each associated prime is precisely the defining property of Cohen-Macaulay modules. The finiteness of $\mathrm{Ass}(M)$, guaranteed automatically for finitely generated modules over Noetherian rings, ensures the local condition determines the global property. □

Lean 4 sketch:

```
theorem cohen_macaulay_of_depth_eq_dim_at_ass
    (R : Type*) [CommRing R] [IsNoetherianRing R]
    (M : Type*) [AddCommGroup M] [Module R M]
    [Module.Finite R M]
    (h : forall p in AssociatedPrimes R M,
         depth R (LocalizedModule p.primeCompl M) =
         krullDim (Localization.AtPrime p)) :
    IsCohenMacaulay R M := sorry
```

Quality: 0.88 Confidence: 0.95 Rigor: 0.95 Judge: publishable

Theorem 4 (Artinian Localizations Imply Weakly Artinian)**.** *Let R be a commutative Noetherian ring and M a finitely generated R-module. If for every $\mathfrak{p} \in \mathrm{Ass}_R(M)$, the localization $M_{\mathfrak{p}}$ is Artinian, then M is weakly Artinian.*

Proof sketch. Since $\mathrm{Ass}(M)$ is finite, the Artinian property at each associated prime ensures the annihilator structure controls the submodule lattice globally. The ascending chain condition on annihilators translates to the weak Artinian condition on M. □

Lean 4 sketch:

```
theorem weakly_artinian_of_artinian_localizations
    (R : Type*) [CommRing R] [IsNoetherianRing R]
    (M : Type*) [AddCommGroup M] [Module R M]
    [Module.Finite R M]
    (h : forall p in AssociatedPrimes R M,
         IsArtinian R (LocalizedModule p.primeCompl M)) :
    IsWeaklyArtinian R M := sorry
```

Quality: 0.75 Confidence: 0.90 Rigor: 0.85

Theorem 5 (Maximal Associated Primes with Bounded Dimension)**.** *Let R be a commutative Noetherian ring and M a finitely generated R-module. If every associated prime of M is maximal and $\dim_R(M) \leq 1$, then M is weakly Artinian.*

Proof sketch. By induction on $\dim M$. Base case ($\dim M = 0$): all associated primes maximal implies M is Artinian by the Auslander–Buchsbaum theorem. Inductive step ($\dim M = 1$): M decomposes into modules R/P^m for maximal P; the dimension constraint bounds the exponents, and localization at each P gives an Artinian module. □

Lean 4 sketch:

```
theorem weakly_artinian_of_max_ass_dim_le_one
    (R : Type*) [CommRing R] [IsNoetherianRing R]
    (M : Type*) [AddCommGroup M] [Module R M]
    [Module.Finite R M]
    (h_max : forall p in AssociatedPrimes R M,
             p.IsMaximal)
    (h_dim : Module.krullDim R M <= 1) :
    IsWeaklyArtinian R M := sorry
```

Quality: 0.78 Confidence: 0.90 Rigor: 0.85 Judge: publishable

Theorem 6 (Artinian at Maximal Associated Primes)**.** *Let R be a commutative Noetherian ring and M a finitely generated R-module. If for every maximal ideal $\mathfrak{m} \in \mathrm{Ass}_R(M)$, the localization $M_{\mathfrak{m}}$ is Artinian, then M is weakly Artinian.*

Proof sketch. The finiteness of $\mathrm{Ass}(M)$ ensures the Artinian property at all maximal associated primes forces M to satisfy the DCC on submodules globally, since the submodule structure is determined by behavior at associated primes. □

Lean 4 sketch:

```
theorem weakly_artinian_of_artinian_at_max_ass
    (R : Type*) [CommRing R] [IsNoetherianRing R]
    (M : Type*) [AddCommGroup M] [Module R M]
```

```
    [Module.Finite R M]
    (h : forall m in AssociatedPrimes R M,
         m.IsMaximal ->
         IsArtinian R (LocalizedModule m.primeCompl M)) :
    IsWeaklyArtinian R M := sorry
```

Quality: 0.80 Confidence: 0.95 Rigor: 0.90 Judge: publishable

A.3 Probability & Statistics

Theorem 7 (Strict Positivity and Conditional Probability)**.** *Let $\mathcal{V}$ be a finite set of random variables with a strictly positive probability distribution. For any disjoint $\mathcal{X}, \mathcal{Z} \subseteq \mathcal{V}$ and $x \in \operatorname{supp}(\mathcal{X})$, $z \in \operatorname{supp}(\mathcal{Z})$:*

$$\Pr(\mathcal{X}{=}x \mid \mathcal{Z}{=}z) > 0 \iff \Pr(\mathcal{X}{=}x, \mathcal{Z}{=}z) > 0.$$

Proof sketch. ($\Rightarrow$) $\Pr(X{=}x, Z{=}z) > 0 \implies \Pr(X{=}x \mid Z{=}z) = \Pr(X{=}x, Z{=}z)/\Pr(Z{=}z) > 0$ since $\Pr(Z{=}z) > 0$ by strict positivity. ($\Leftarrow$) $\Pr(X{=}x \mid Z{=}z) > 0 \implies \Pr(X{=}x, Z{=}z) = \Pr(X{=}x \mid Z{=}z) \cdot \Pr(Z{=}z) > 0$. □

Lean 4 sketch:

```
theorem strict_positivity_conditional
    {V : Type*} [Fintype V]
    (P : ProbabilityMeasure V) (hP : StrictlyPositive P)
    (X Z : Set V) (hXZ : Disjoint X Z)
    (x : V) (hx : x in X) (z : V) (hz : z in Z) :
    P.cond X Z x z > 0 <-> P.joint X Z x z > 0 := sorry
```

Quality: 0.60 Confidence: 0.95 Rigor: 0.90

Theorem 8 (Conditional Independence Implies Marginal Independence)**.** *Let $\mathcal{V}$ be a finite set of discrete random variables with a strictly positive distribution. For any disjoint $\mathcal{X}, \mathcal{Z} \subseteq \mathcal{V}$, if $\mathcal{X} \perp\!\!\!\perp \mathcal{Z} \mid \mathcal{Y}$ for some $\mathcal{Y} \subseteq \mathcal{V}$, then $\mathcal{X} \perp\!\!\!\perp \mathcal{Z}$.*

Proof sketch. Under strict positivity, conditional independence gives $P(X, Z \mid Y{=}y) = P(X \mid Y{=}y)\, P(Z \mid Y{=}y)$ for all y. Summing: $P(X, Z) = \sum_y P(X, Z \mid Y{=}y)\, P(Y{=}y) = \sum_y P(X \mid Y{=}y)\, P(Z \mid Y{=}y)\, P(Y{=}y) = P(X)\, P(Z)$. □

Lean 4 sketch:

```
theorem cond_indep_implies_marginal_indep
    {V : Type*} [Fintype V] [DecidableEq V]
    (P : ProbabilityMeasure V) (hP : StrictlyPositive P)
    (X Z Y : Set V) (hXZ : Disjoint X Z)
    (h_ci : CondIndep X Z Y P) :
    Indep X Z P := sorry
```

Quality: 0.78 Confidence: 0.95 Rigor: 0.90 Judge: publishable

Theorem 9 (Concentration of Sample Second Moment)**.** *Let $X_1, \ldots, X_n$ be i.i.d. random variables with finite second moment. Suppose $\exists\, \beta \in (0, 1]$ such that $\forall\, t > 0$: $\Pr[\max_{1 \le i \le n} |X_i| > t] \le e^{-\beta t^2}$. Then $\forall\, \varepsilon > 0$, $\exists\, C > 0$ such that*

$$\Pr\left[\left|\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \mathbb{E}[X_1^2]\right| > \varepsilon\right] \le C\, e^{-n\beta\varepsilon^2/4}.$$

Proof sketch. Apply a Bernstein-type concentration inequality. The tail bound $\Pr[\max |X_i| > t] \leq e^{-\beta t^2}$ controls the moment generating function of X_i^2, yielding exponential concentration of $n^{-1} \sum X_i^2$ around $\mathbb{E}[X_1^2]$. □

Lean 4 sketch:

```
theorem sample_variance_concentration
    (n : N) (beta eps : R)
    (hb : 0 < beta) (hb1 : beta <= 1) (he : 0 < eps)
    (X : Fin n -> Omega -> R) (hX_iid : IID X)
    (h_tail : forall t > 0,
      Prob (max_i |X i| > t) <= exp (-beta * t^2)) :
    exists C > 0,
      Prob (|mean (fun i => (X i)^2) -
            E[(X 0)^2]| > eps) <=
      C * exp (-n * beta * eps^2 / 4) := sorry
```

Quality: 0.76 Confidence: 0.90 Rigor: 0.95 Judge: publishable